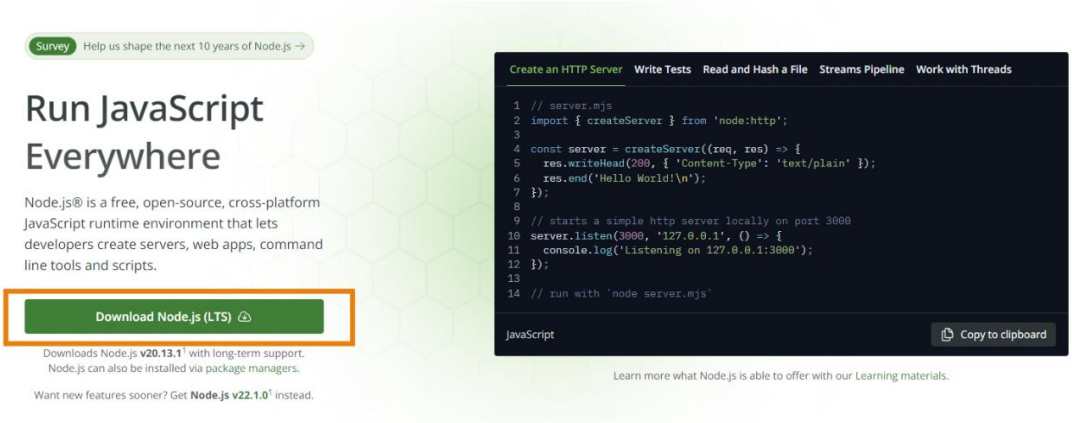
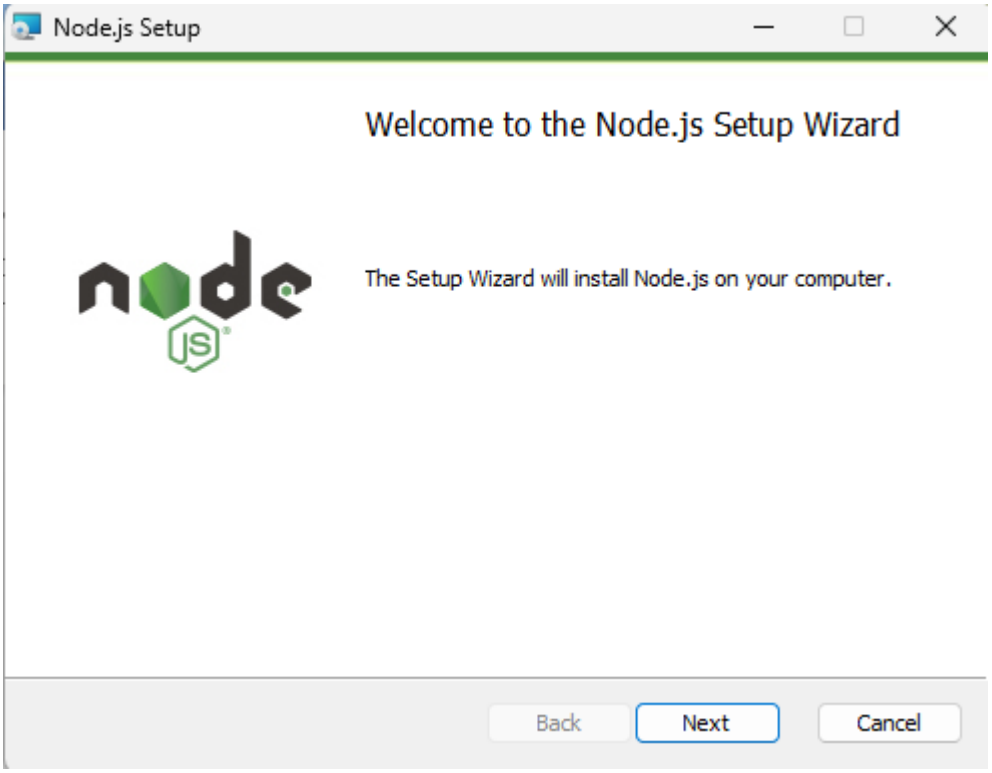
使用軟體：node.js、MySQL、Python
MySQL 為本專案使用的資料庫系統，請依自身需求決定使用的資料庫系統
以下為在 Windows 環境下的架設教學。

Node.js 安裝

1. 請先前往 [nodejs 官網](#) 安裝 nodejs，安裝穩定版（LTS）即可。

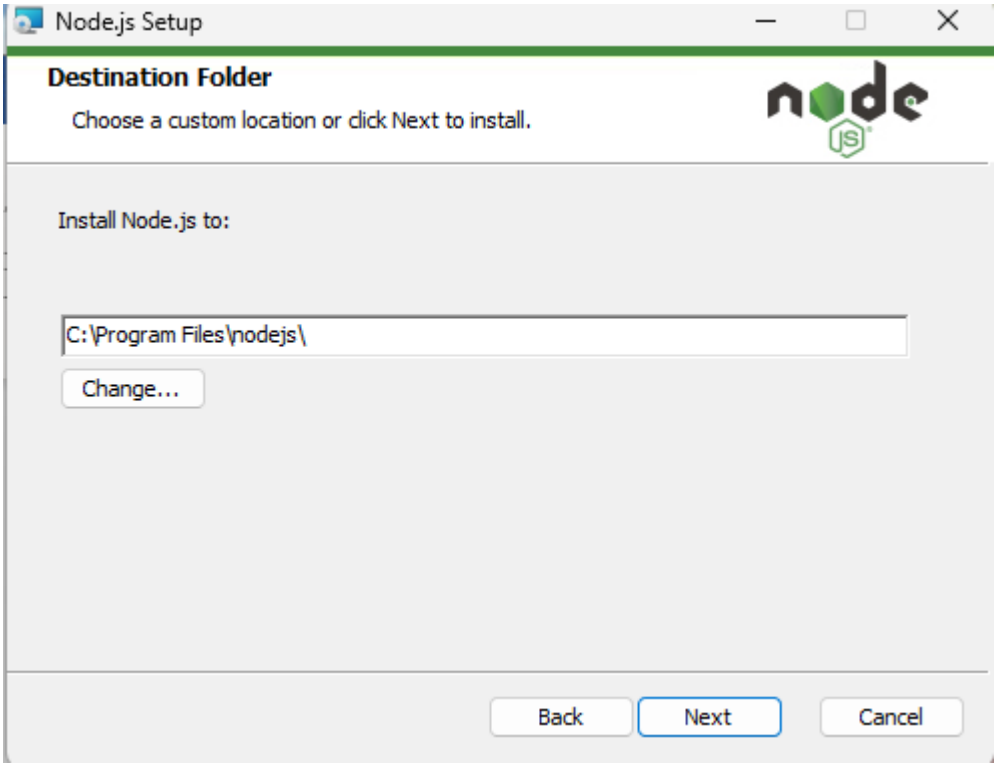


2. 下載完成後，點選安裝檔案後開始安裝。

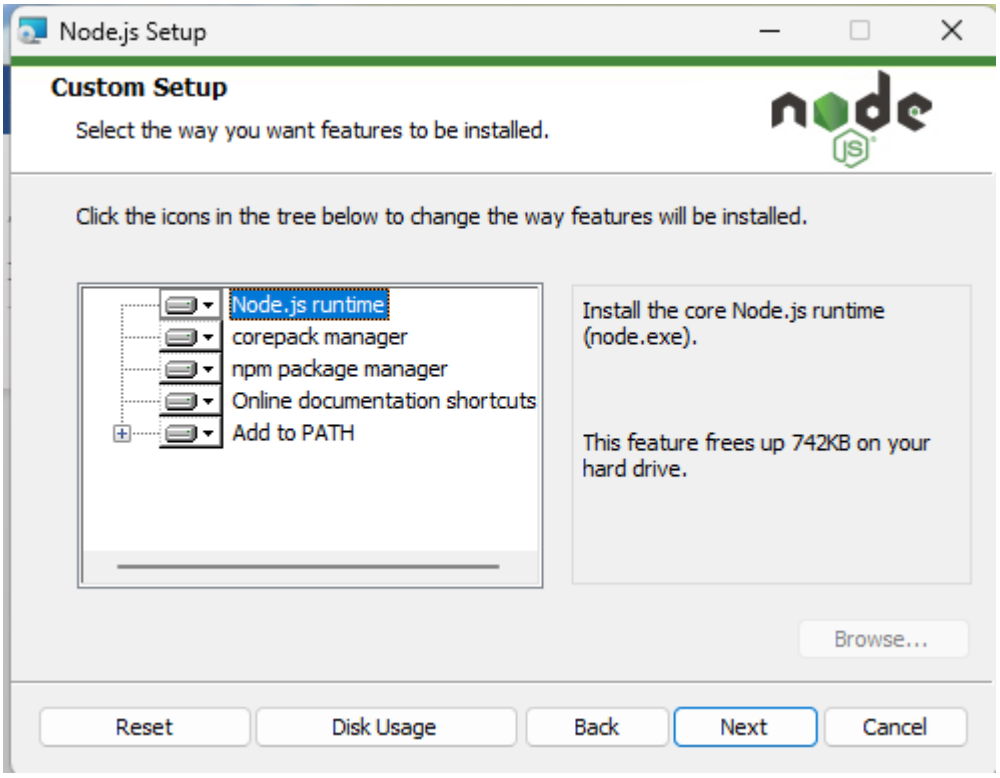


3. End-User License Agreement 的頁面，勾選 "I accept the terms in the License Agreement" 核選方塊後，點選 Next 進行下一步

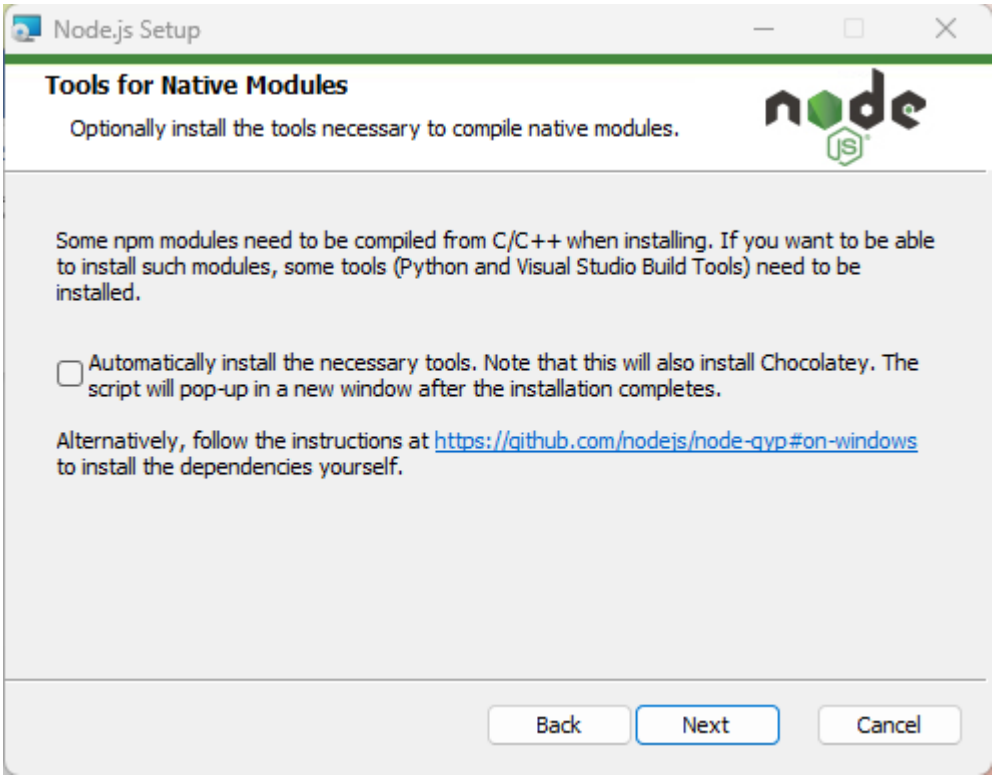
4. 可指定安裝路徑，這裡我保持預設，確認點選 Next。



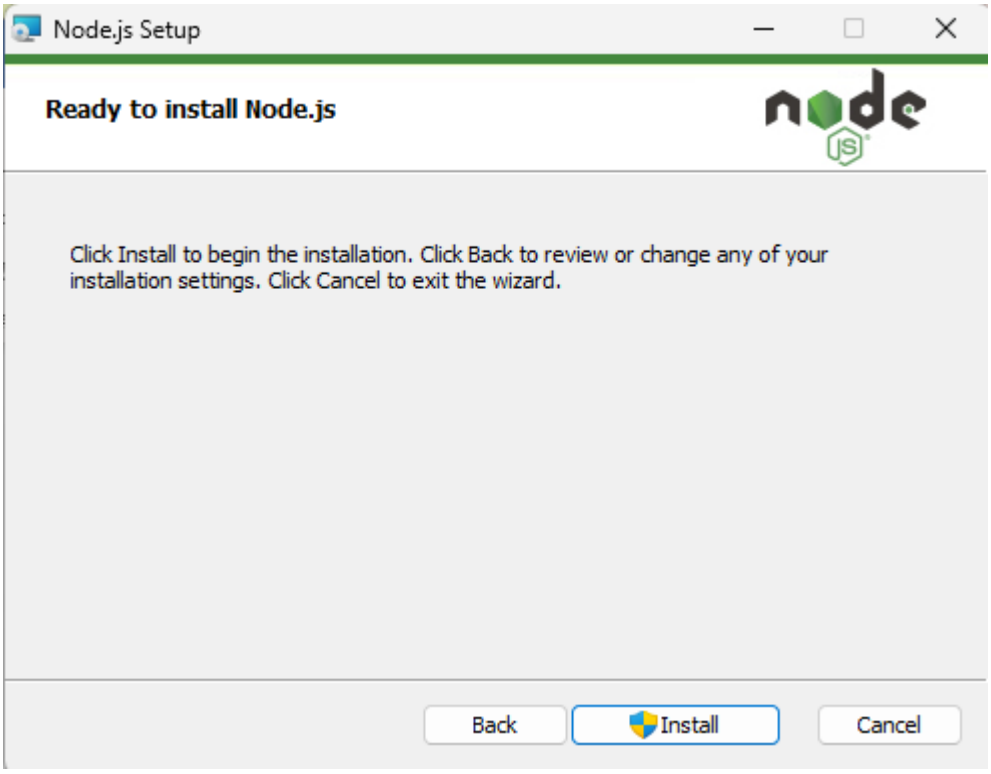
5. 自訂安裝，依需求自行設定，這裡我不做更改直接 Next。



6. 設定是否安裝 Tools for Native Modules，可以自行決定是否勾選，這邊不做勾選，點擊 Next。



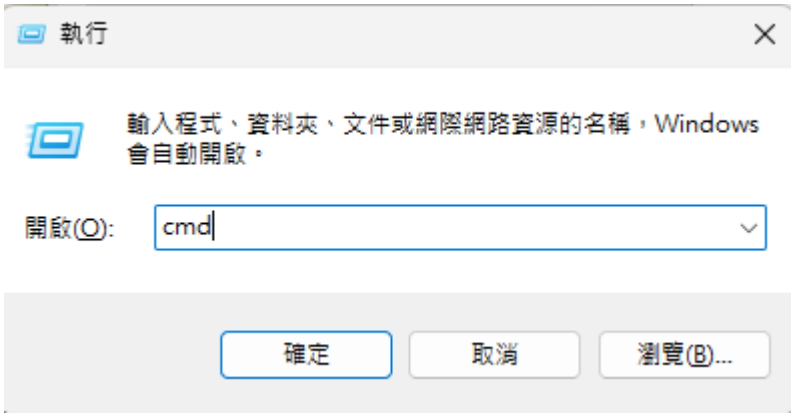
7. 安裝前的提醒，點擊 Install。



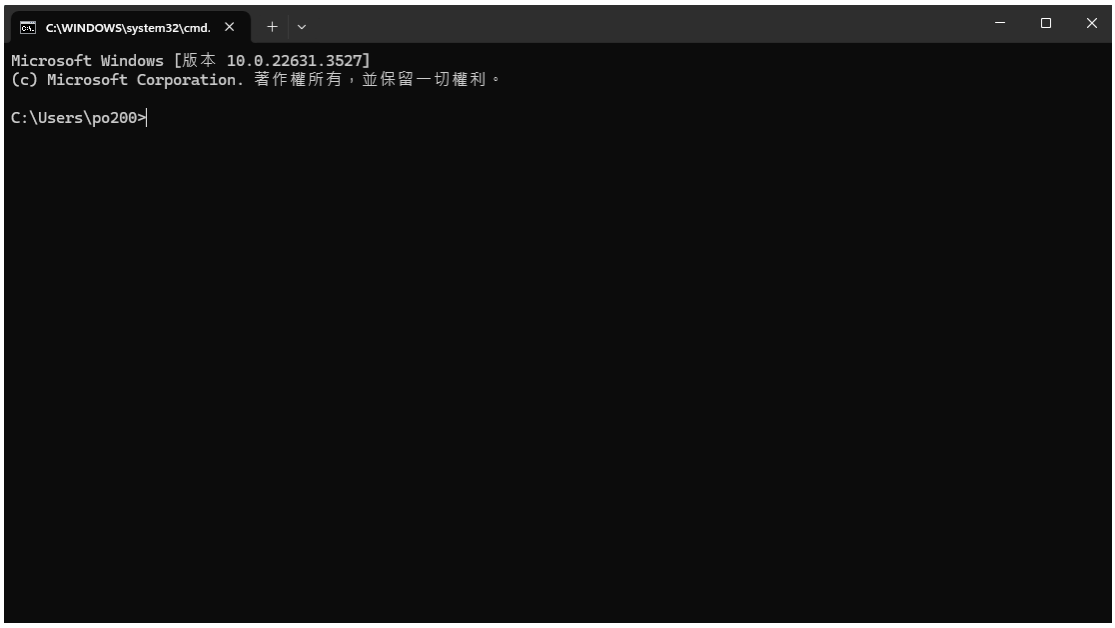
8. 完成安裝

驗證 Node.js 是否正確安裝

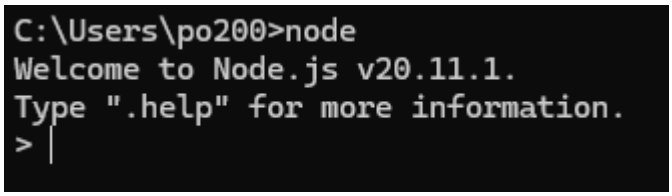
點擊鍵盤上的 Windows 鍵 + R 鍵，會出現下面的視窗。



出現後輸入 cmd 後按下確定，會出現命令提示字元的視窗如下。



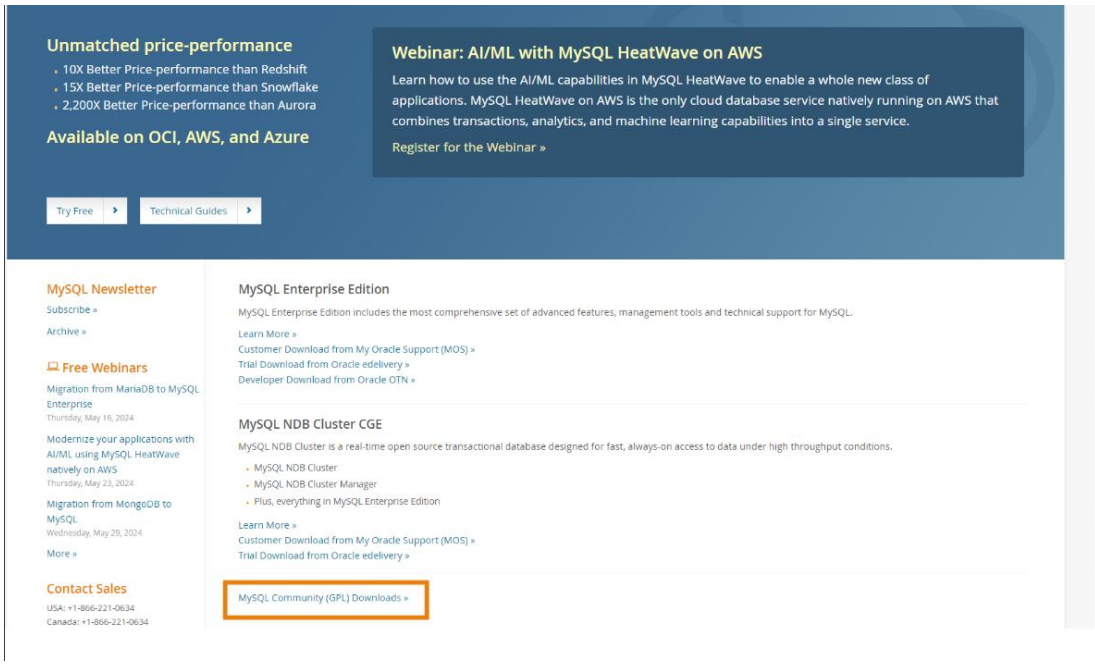
然後輸入 node ，後點擊 Enter 鍵，如果出現類似以下的畫面，就代表安裝成功，如果沒有出現的話，請重新啟動電腦後再次嘗試。



MySQL 安裝

1.前往 MySQL 的官網，到官網後點擊 DOWNLOADS

2.到最下面會看到 [MySQL Community (GPL) Downloads »](#) ，點擊它



3. 我的電腦是 Windows，所以我點選 MySQL Installer for Windows。

4. 這邊依自身需求決定安裝的版本，我這邊是直接安裝官網上的最新版本，這邊有 2 個安裝方式，一個是 Web Community 版(線上安裝)，另一個是 Community 版(離線安裝)，這邊我選擇下面的 Community 版。

MySQL Community Downloads

MySQL Installer

General Availability (GA) Releases | Archives

MySQL Installer 8.0.37

Note: MySQL 8.0 is the final series with MySQL Installer. As of MySQL 8.1, use a MySQL product's MSI or Zip archive for installation. MySQL Server 8.1 and higher also bundle MySQL Configurator, a tool that helps configure MySQL Server.

Select Version:

8.0.37

Select Operating System:

Microsoft Windows

Windows (x86, 32-bit), MSI Installer	8.0.37	2.1M	Download
(mysql-installer-web-community-8.0.37.0.msi)		MD5: 398f1365f2bd43af9f6ece9add565c1b | Signature	
Windows (x86, 32-bit), MSI Installer	8.0.37	296.1M	Download
(mysql-installer-community-8.0.37.0.msi)		MD5: ae605e4f62aaf8bb1adef684d62a49f2 | Signature	

We suggest that you use the MD5 checksums and GnuPG signatures to verify the integrity of the packages you download.

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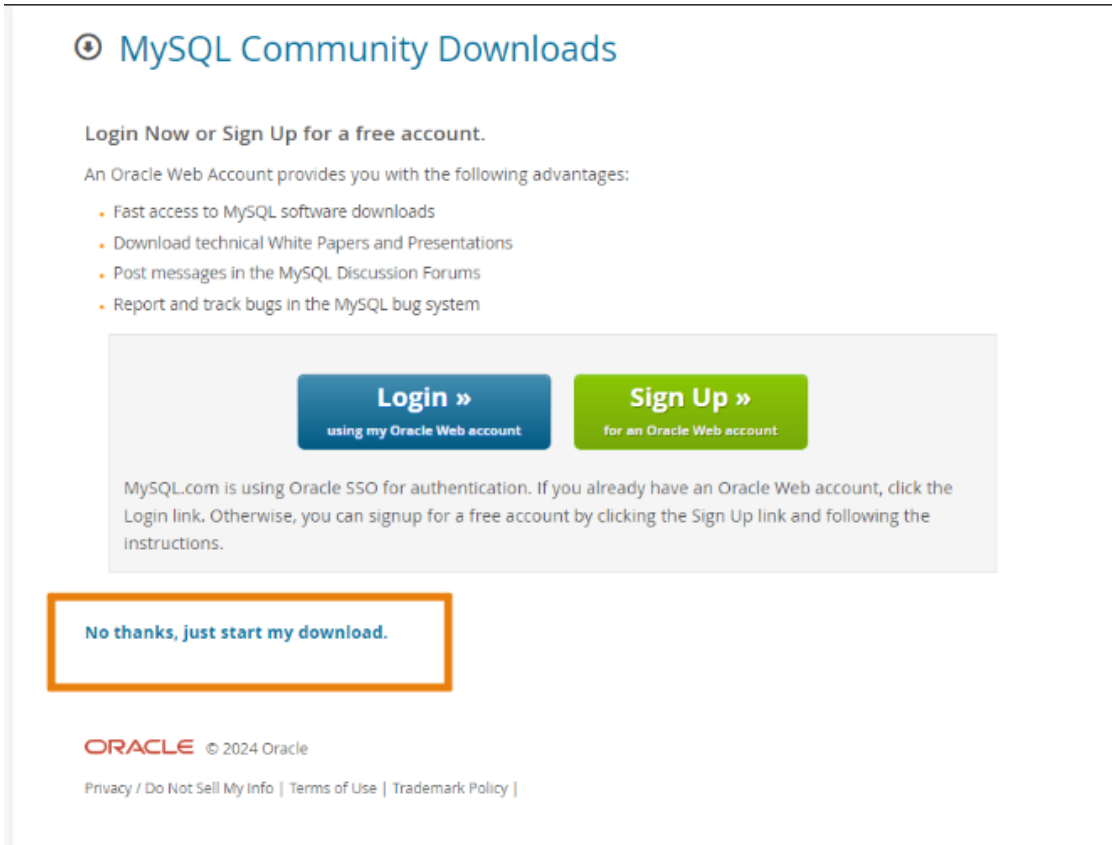
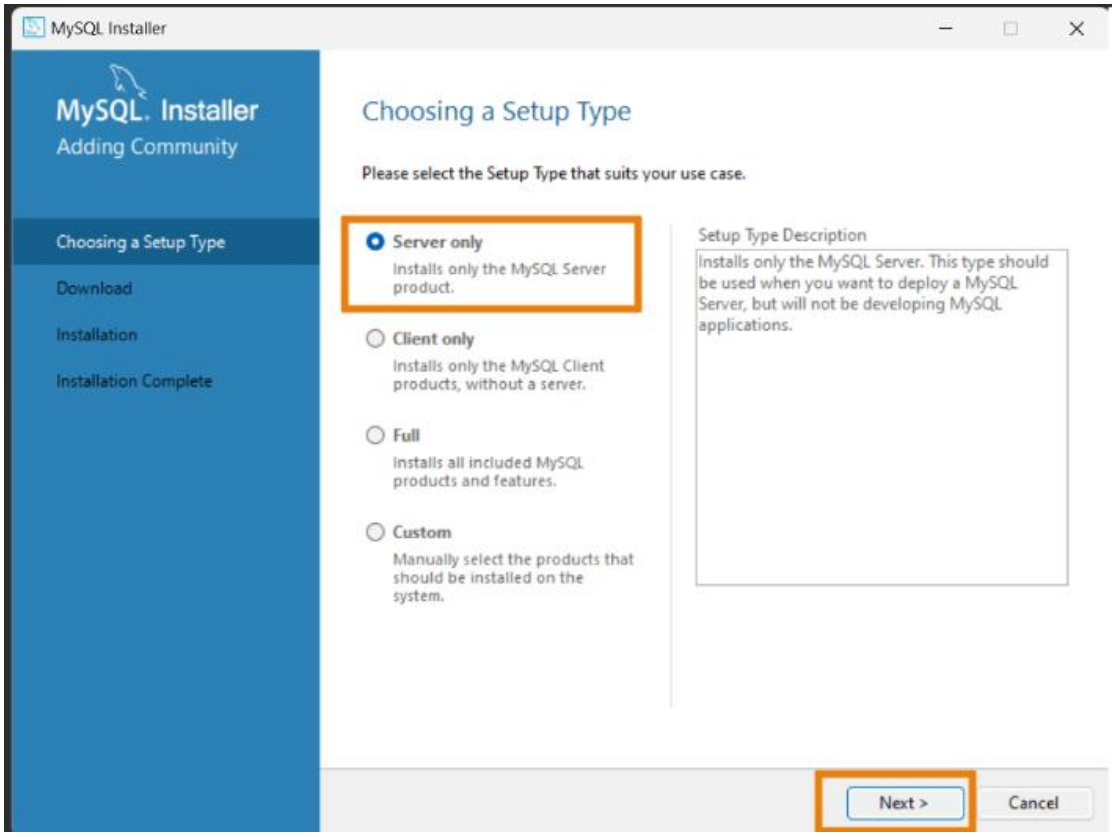
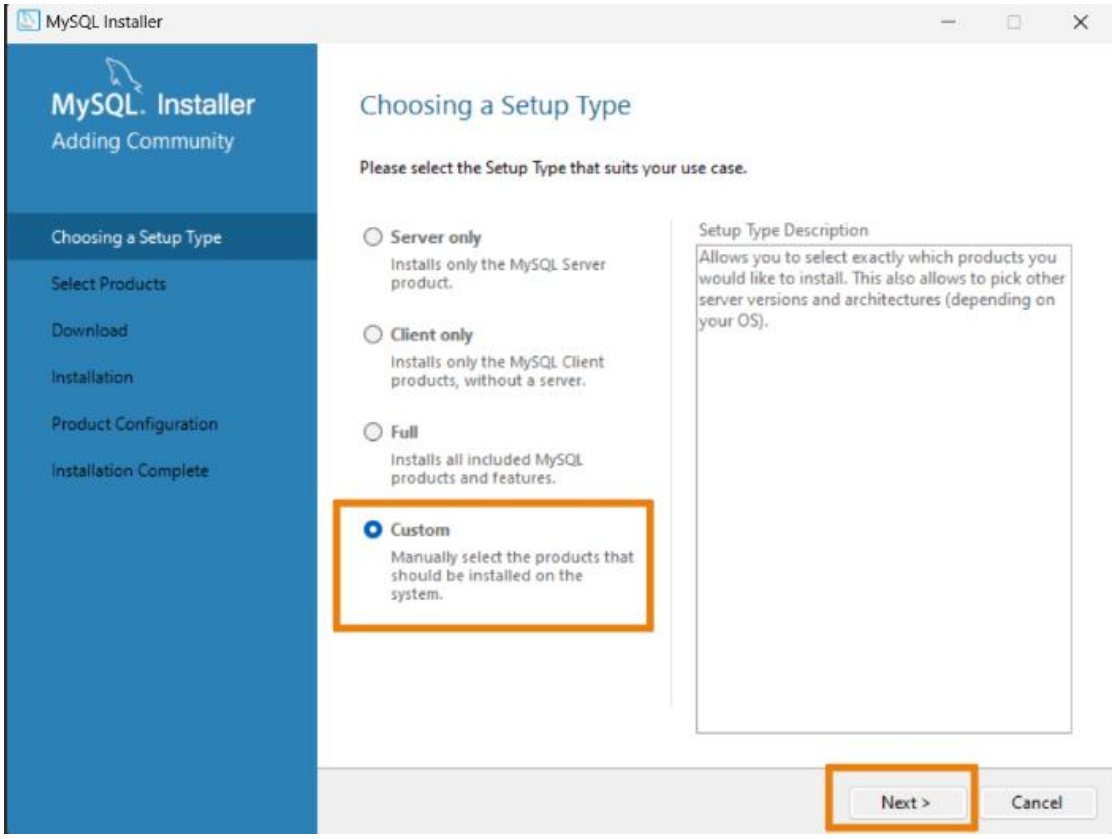


5. 點擊好後，開啟下載到電腦上的安裝檔，接下來就是一連串的 Next 跟 Execute。

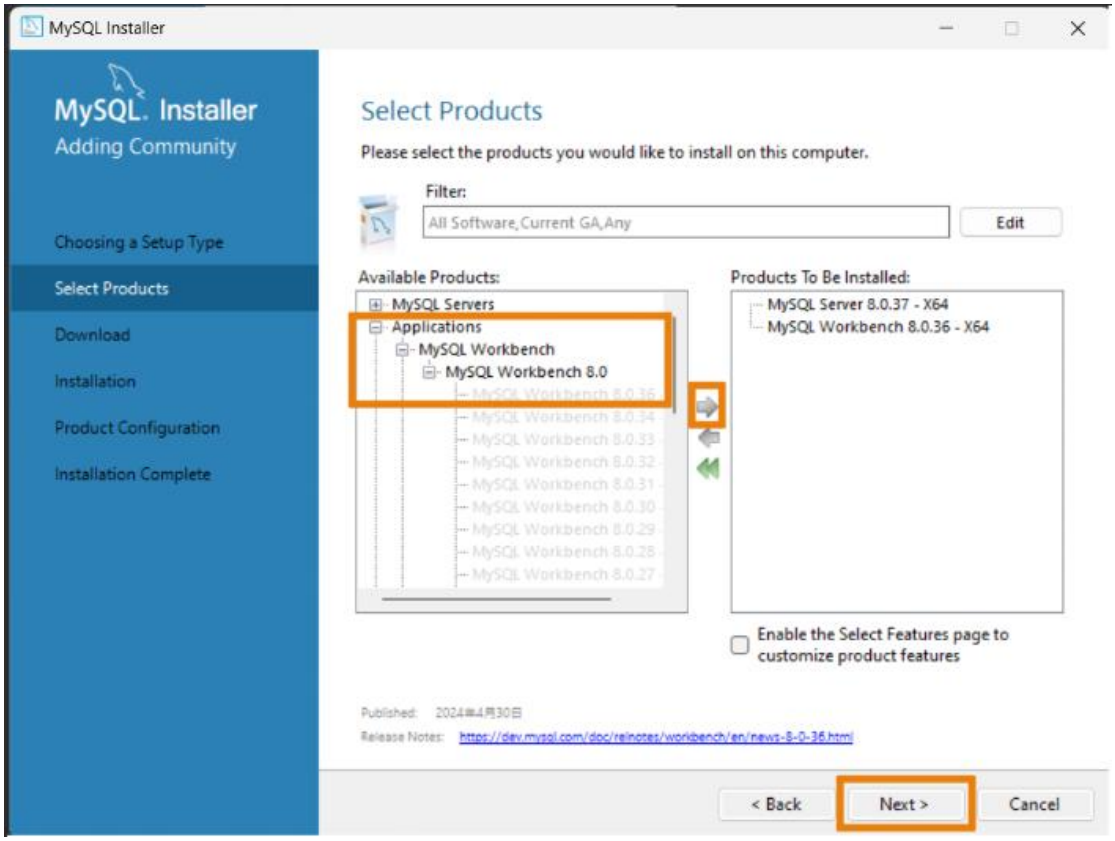
如果不需介面管理資料庫的話，可以選用 Server only。

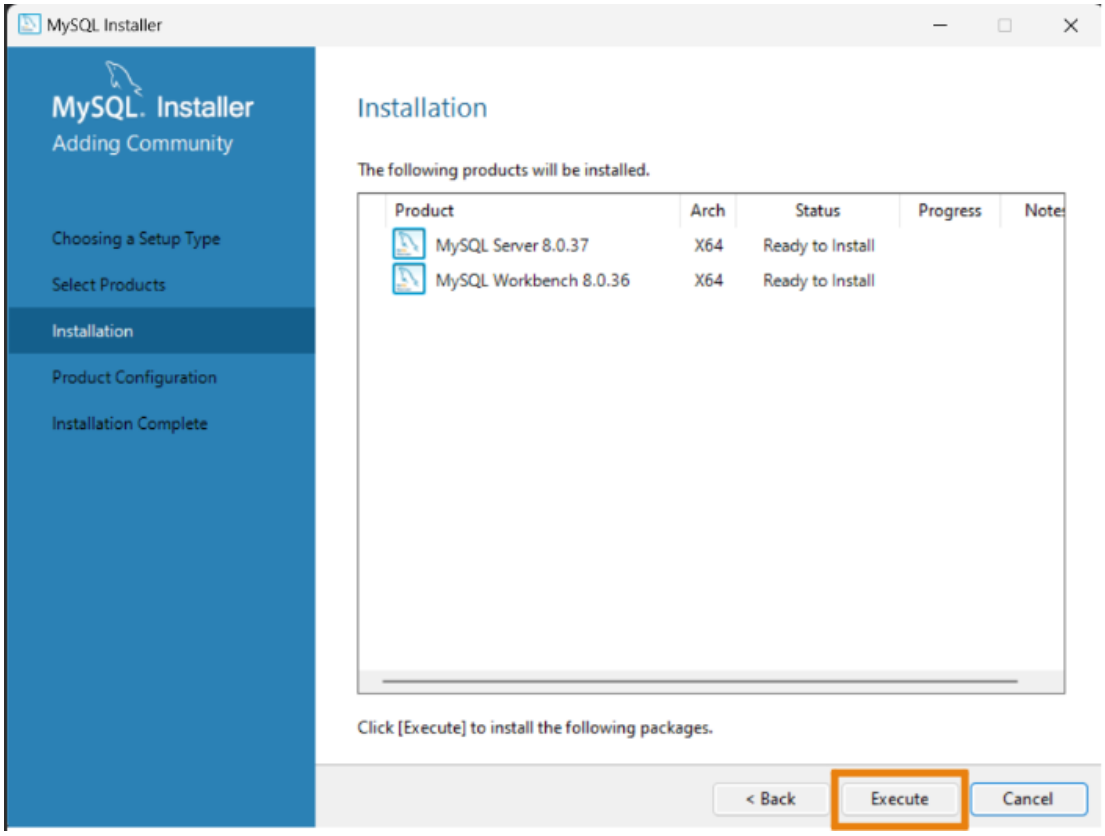


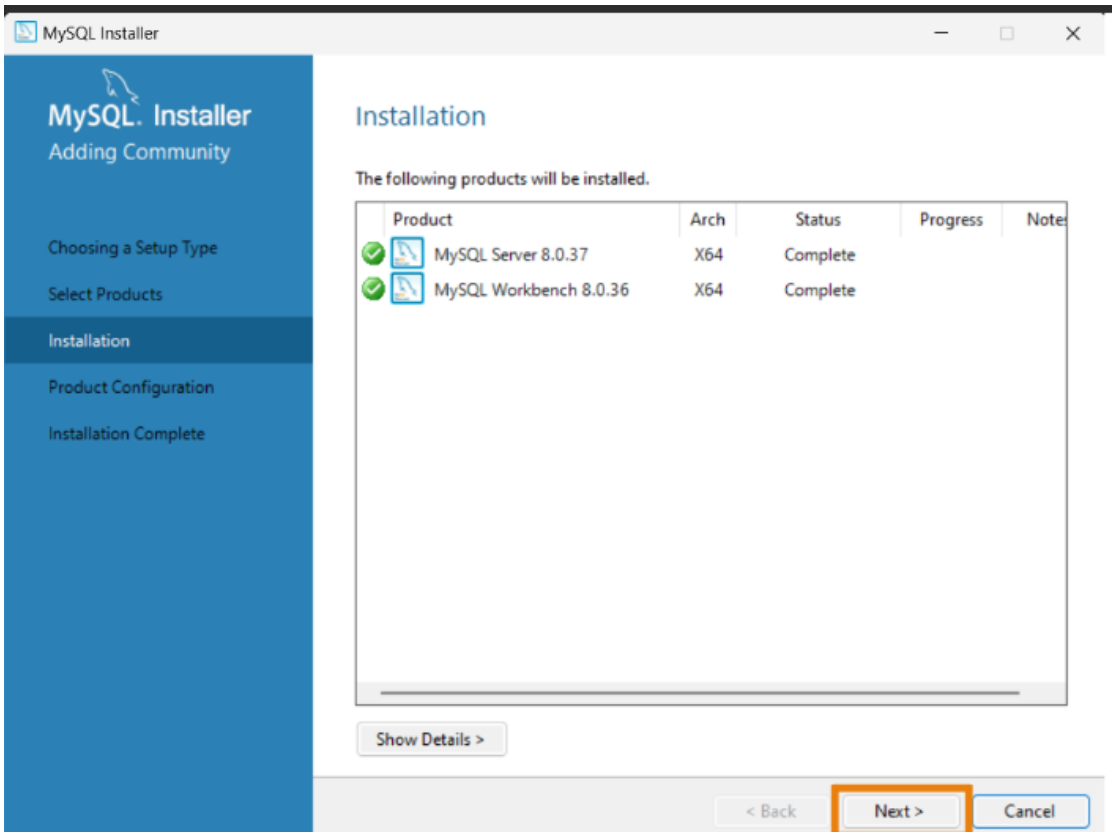
如果要資料庫管理介面的話，請選擇 Custom。

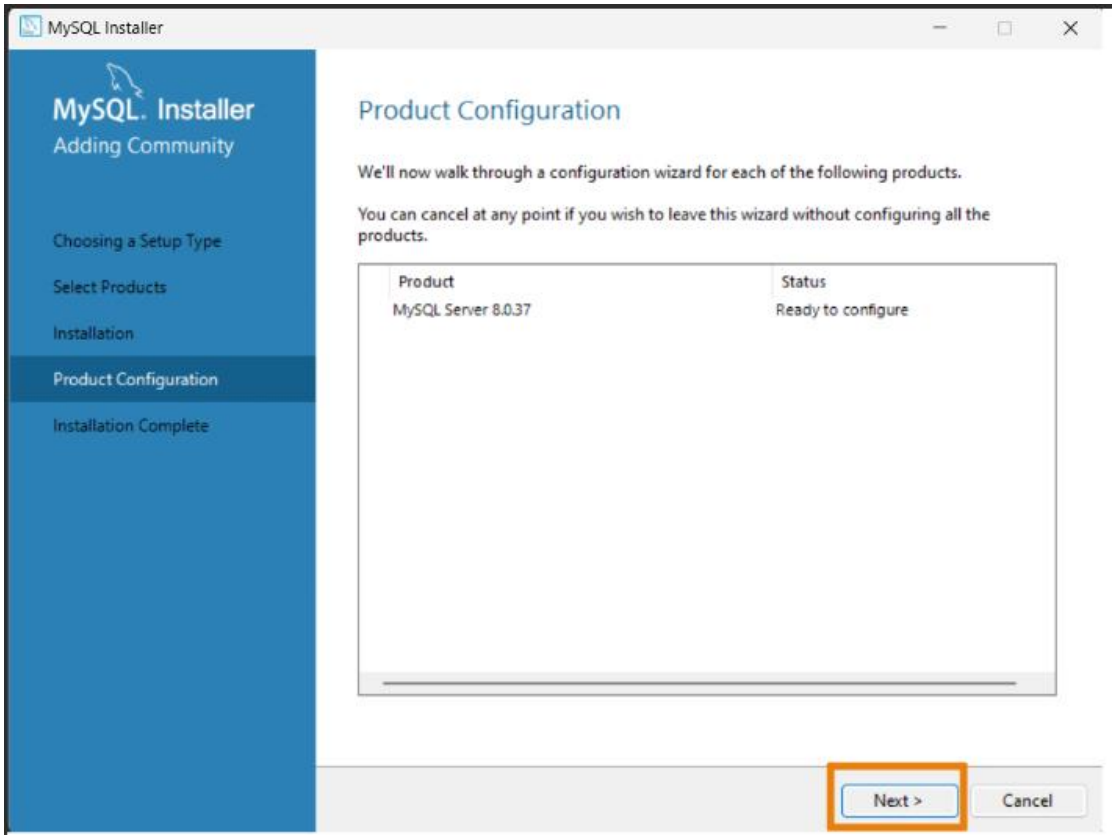


點開 Application 左方的+號，並點選 MySQL Workbench 左方的+號，再點選 MySQL Workbench 8.0 左方的+號，這邊可以選擇想要的版本，這裡用的是 8.0.36 版，點擊版本後，按下中間朝右的箭頭，它就會添加到要安裝的項目了。

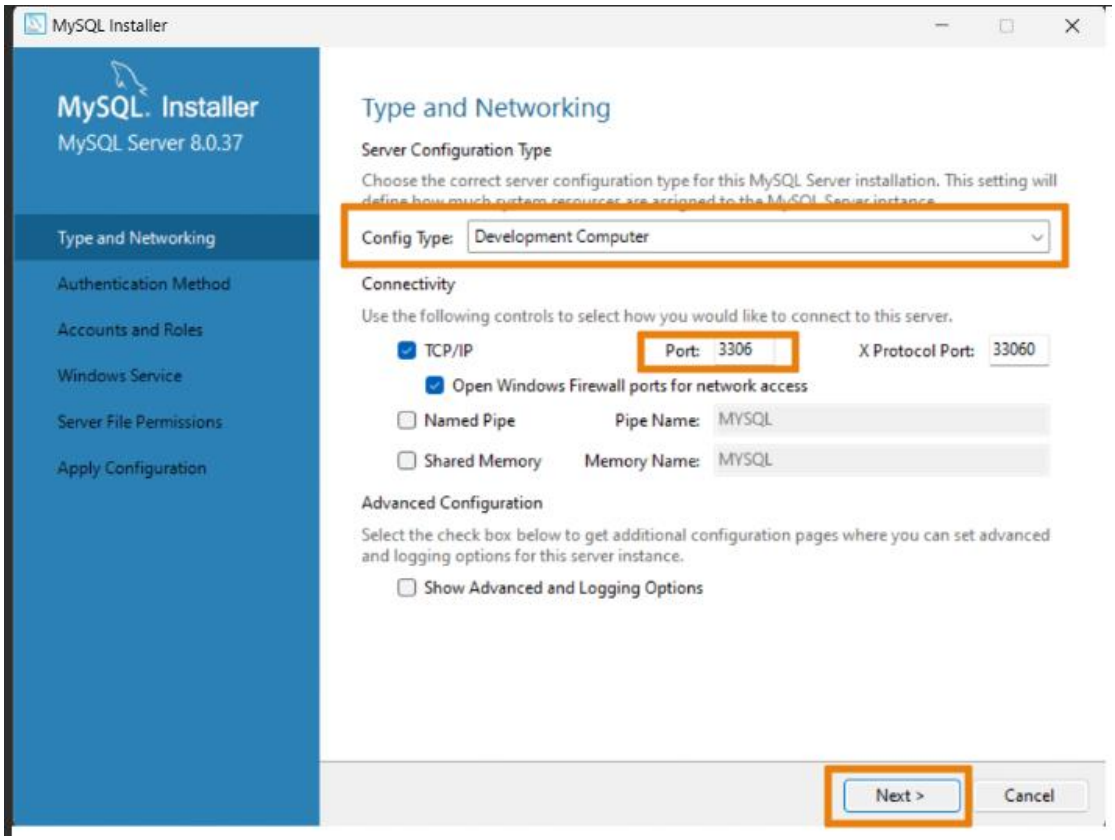

MySQL Installer
MySQL. Installer
Adding Community
Choosing a Setup Type
Select Products
Download
Installation
Product Configuration
Installation Complete
Select Products
Please select the products you would like to install on this computer.
Filter:
All Software,Current GA,Any
Edit
Available Products:
MySQL Servers
Applications
MySQL Workbench
MySQL Workbench 8.0
Products To Be Installed:
MySQL Server 8.0.37 - X64
MySQL Workbench 8.0.36 - X64
Enable the Select Features page to customize product features
< Back
Next >
Cancel


MySQL Installer
MySQL. Installer
Adding Community
Choosing a Setup Type
Select Products
Installation
Product Configuration
Installation Complete
Installation
The following products will be installed.
Product
Arch
Status
Progress
Note
MySQL Server 8.0.37
X64
Ready to Install
MySQL Workbench 8.0.36
X64
Ready to Install
Click [Execute] to install the following packages.
< Back
Execute
Cancel

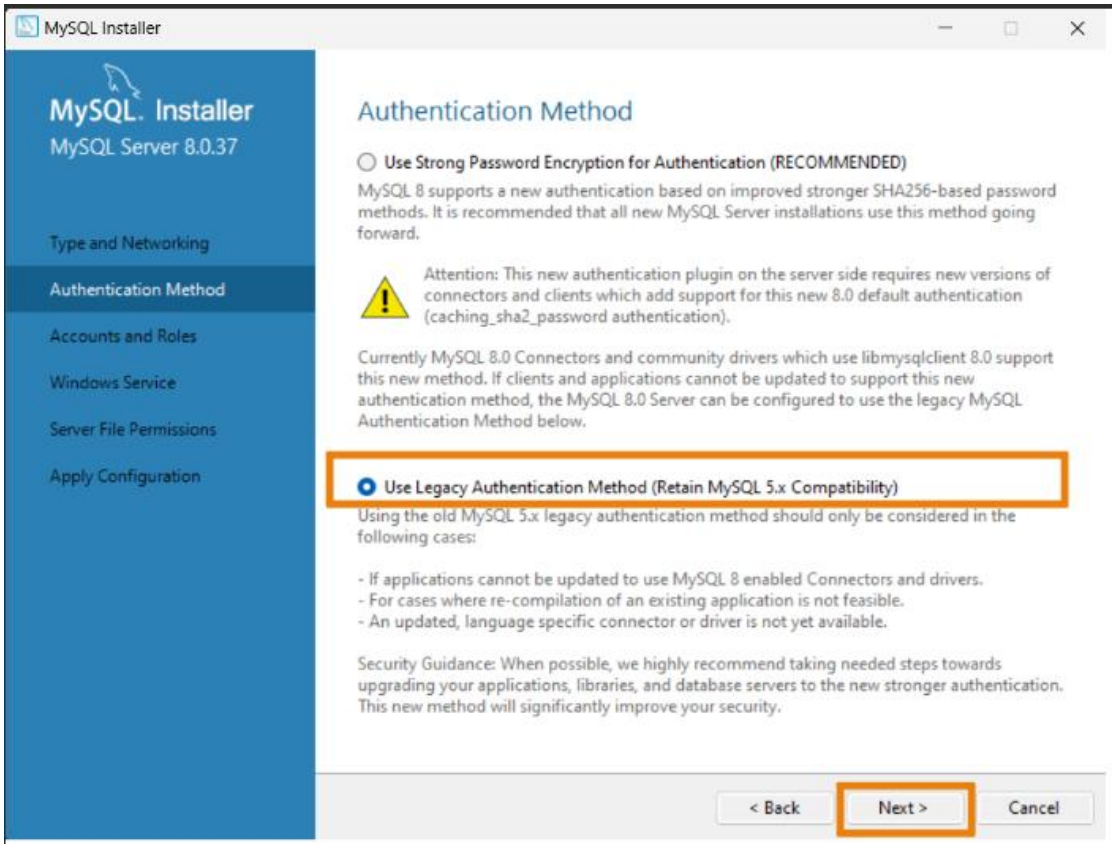




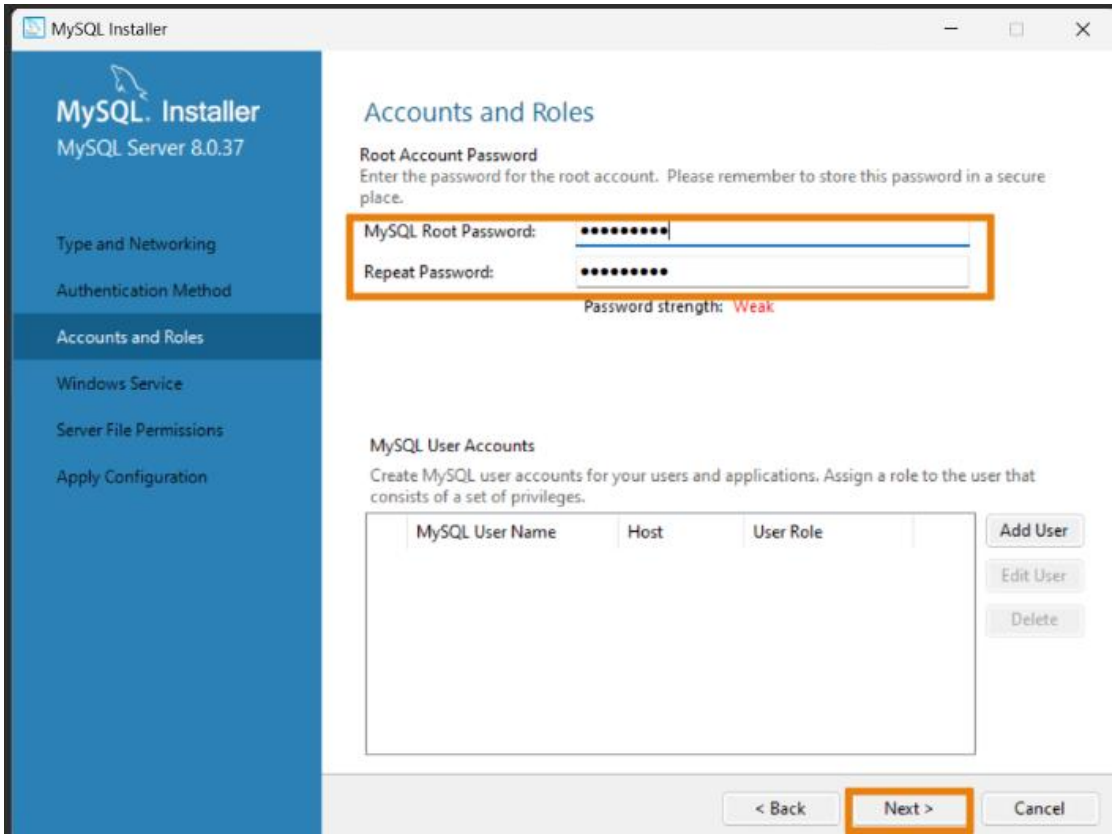
請依資料量和專案大小自行決定 Config Type，由於本專案測試的資料量不大，所以使用預設的 Development Computer，Port 使用預設的 3306 就可以了。



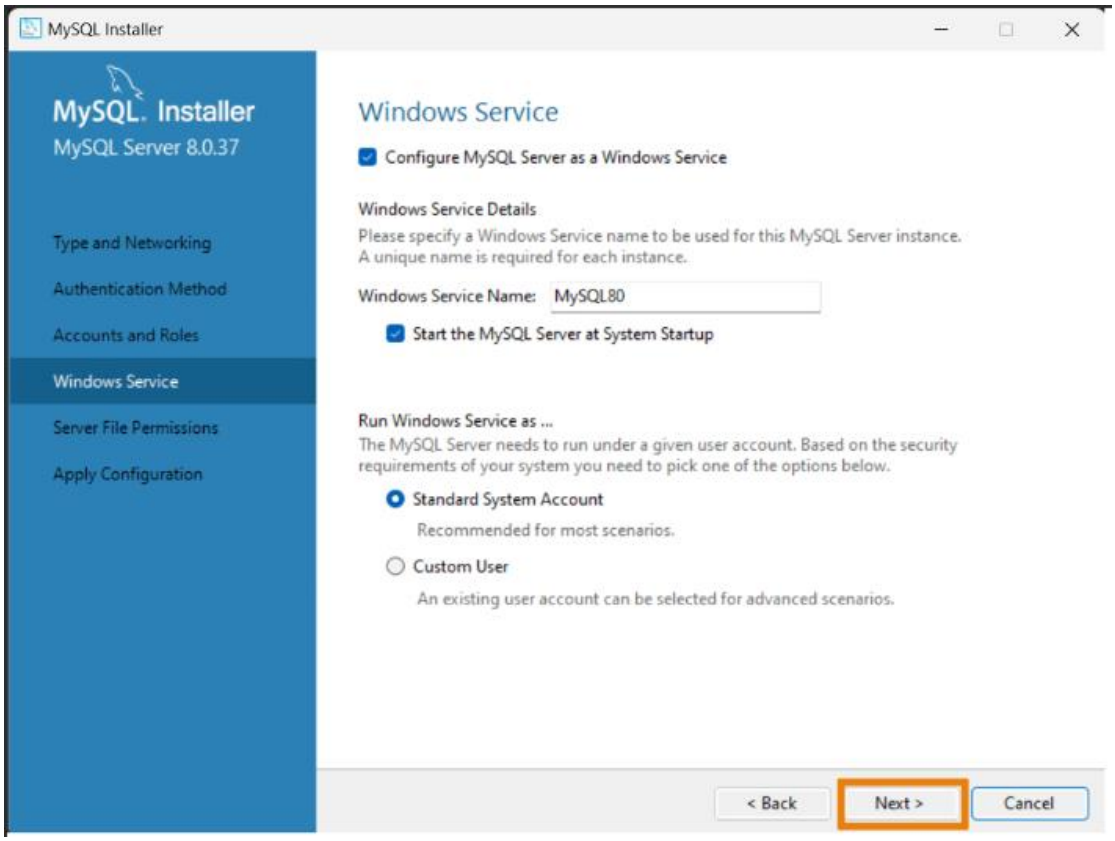
由於 node.js 的 db 套件不支援 caching_sha2_password，所以選擇下面的 Use Legacy Authentication Method。

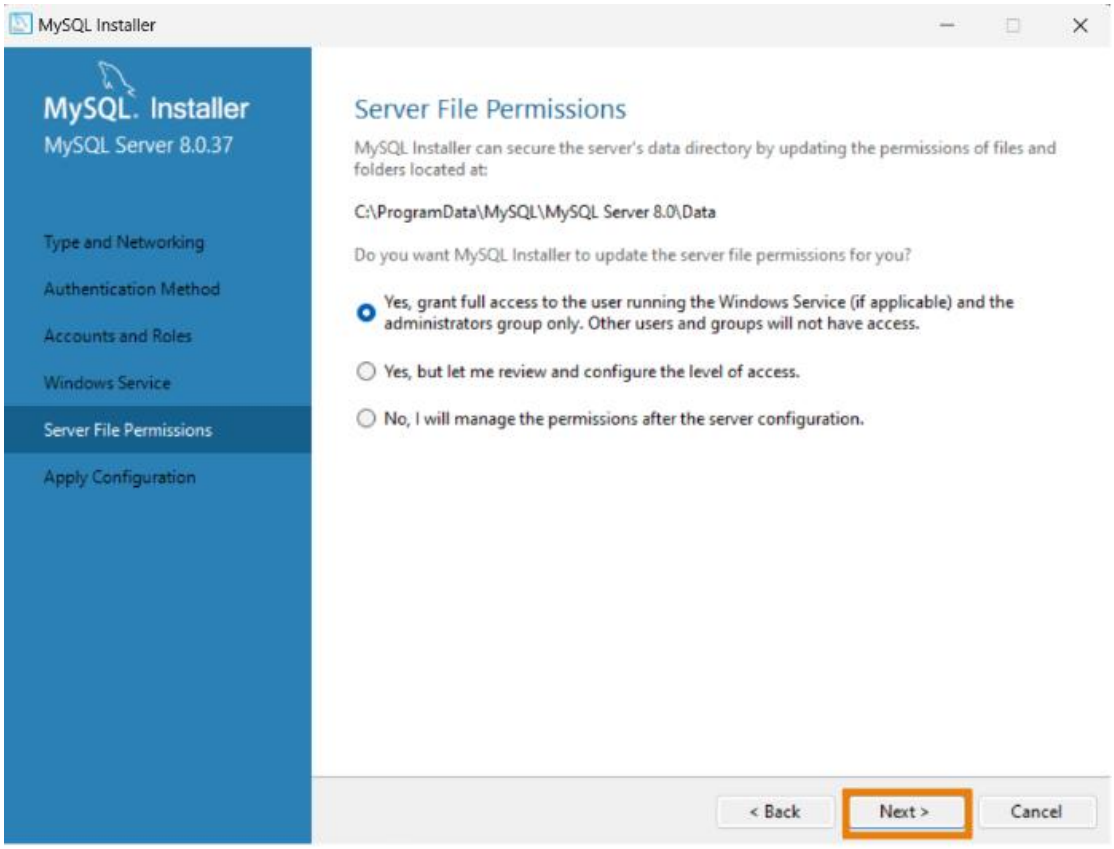


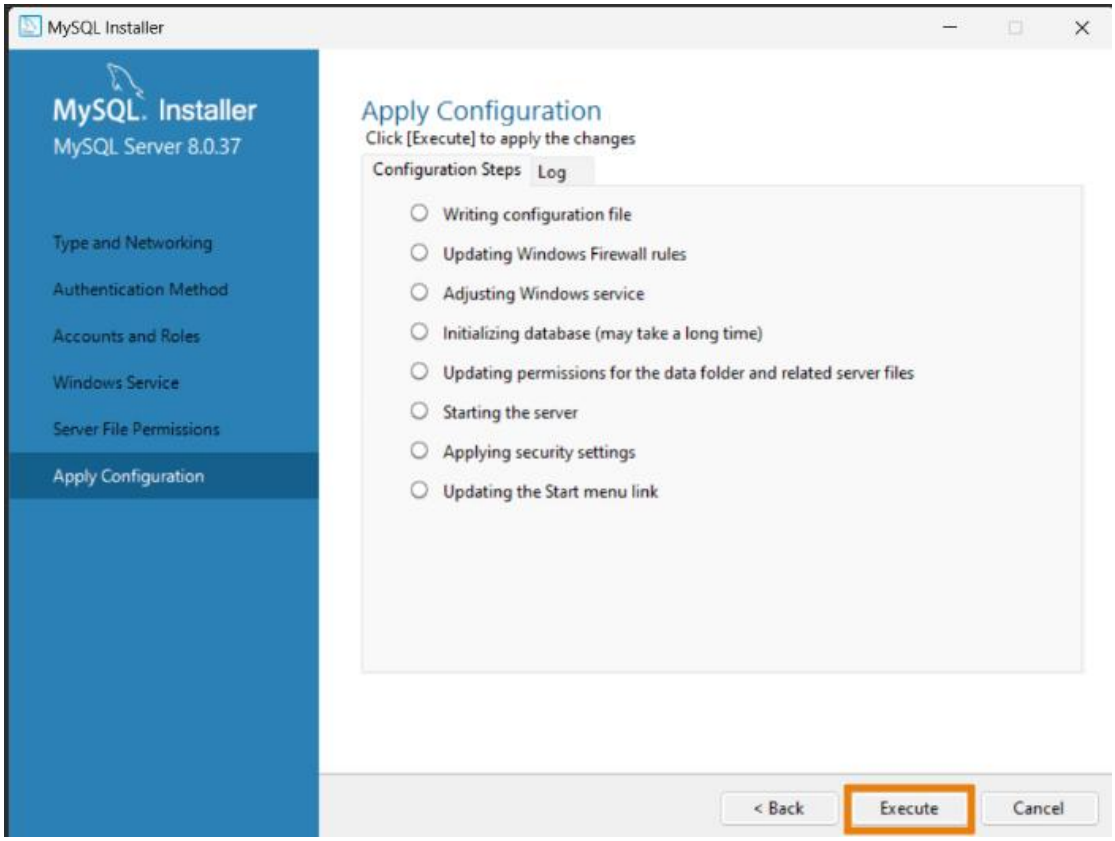
這是設定自己 MySQL 的 root 帳號的密碼，後續連接 MySQL 會用到。

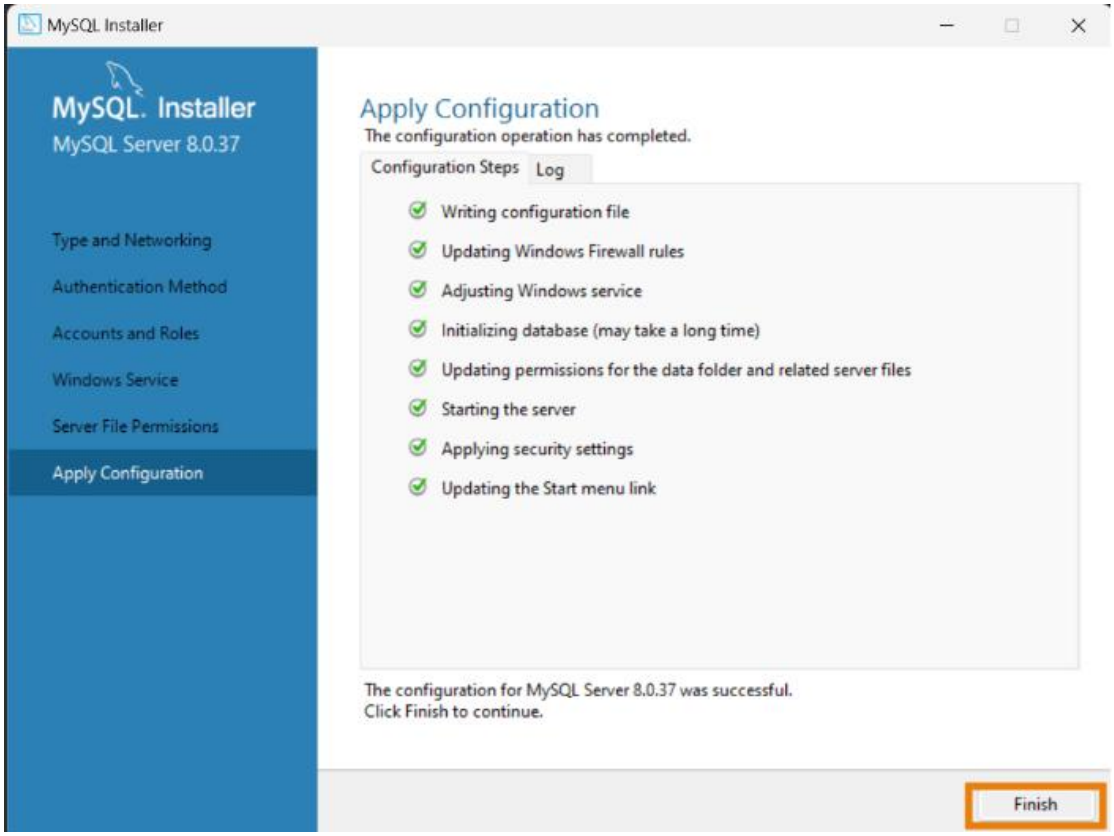


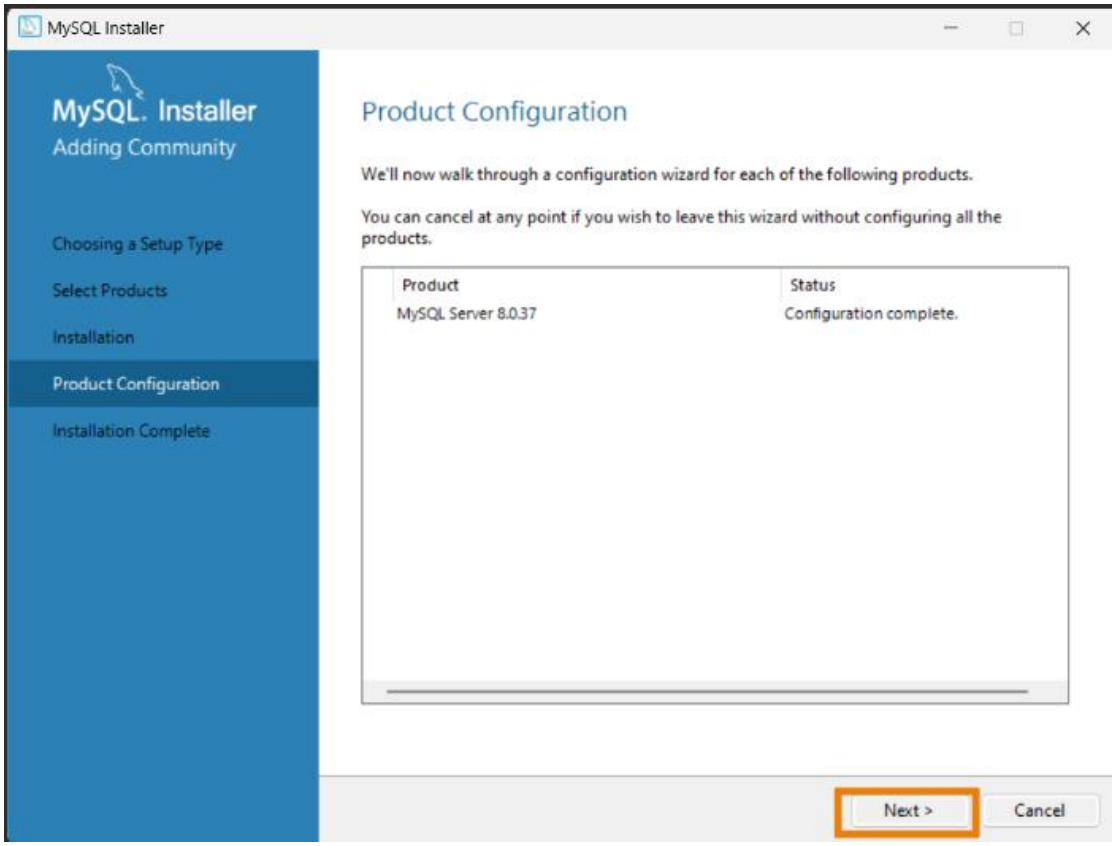
依自身需求決定是否更改 Windwos Service Name，這邊我使用預設名稱。

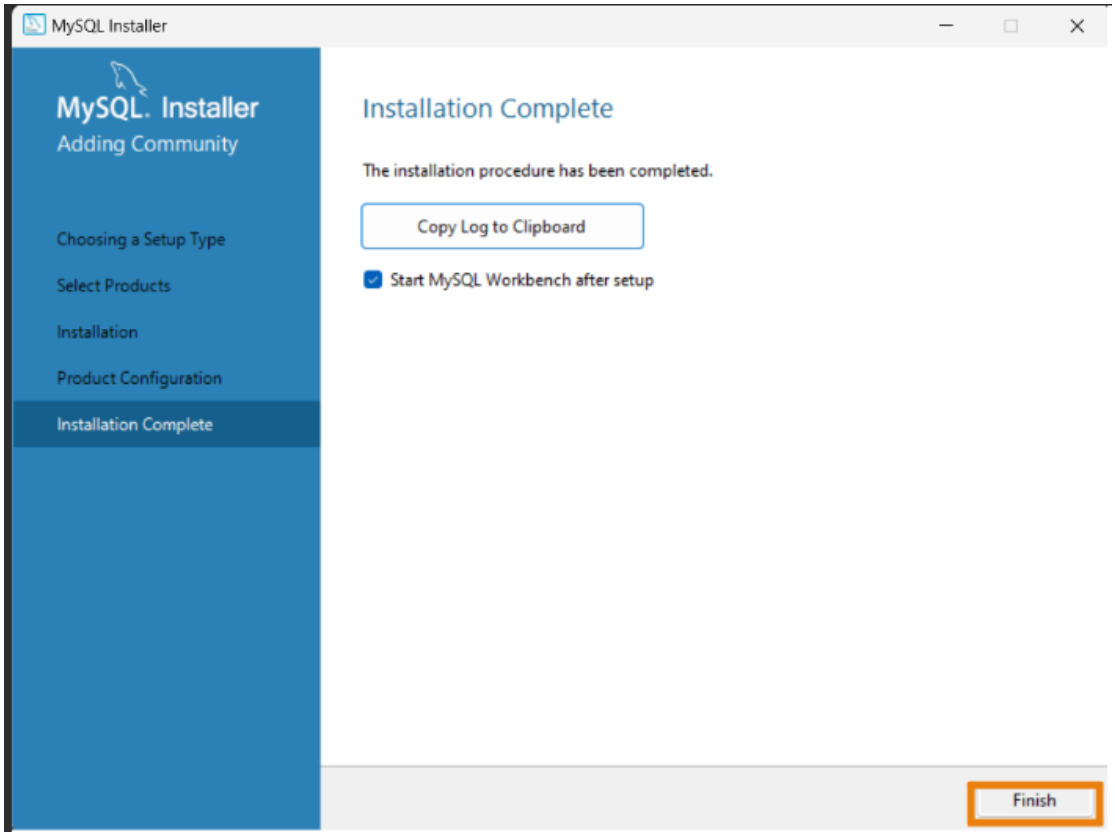



MySQL Installer
MySQL. Installer
MySQL Server 8.0.37
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration
Server File Permissions
MySQL Installer can secure the server's data directory by updating the permissions of files and folders located at:
C:\ProgramData\MySQL\MySQL Server 8.0\Data
Do you want MySQL Installer to update the server file permissions for you?
Yes, grant full access to the user running the Windows Service (if applicable) and the administrators group only. Other users and groups will not have access.
Yes, but let me review and configure the level of access.
No, I will manage the permissions after the server configuration.
< Back
Next >
Cancel


MySQL Installer
MySQL. Installer
MySQL Server 8.0.37
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration
Apply Configuration
Click [Execute] to apply the changes
Configuration Steps
Log
Writing configuration file
Updating Windows Firewall rules
Adjusting Windows service
Initializing database (may take a long time)
Updating permissions for the data folder and related server files
Starting the server
Applying security settings
Updating the Start menu link
< Back
Execute
Cancel

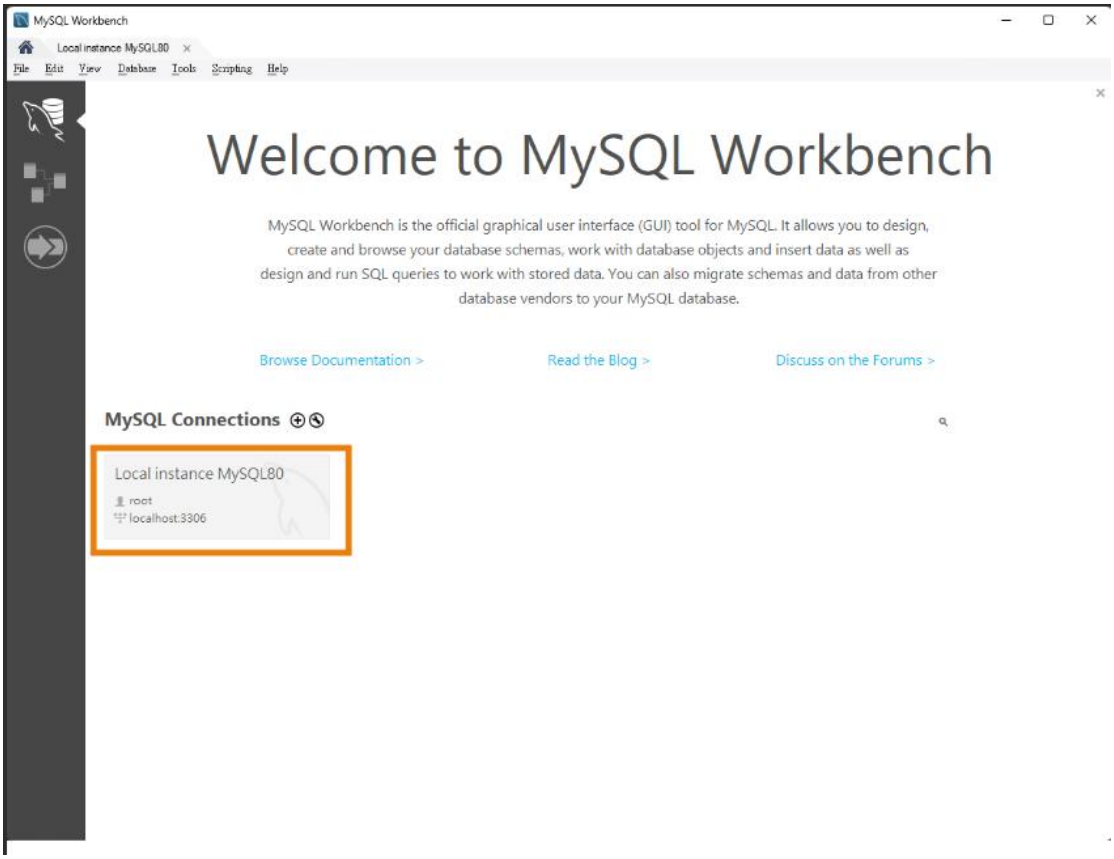

MySQL Installer
MySQL. Installer
MySQL Server 8.0.37
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration
Apply Configuration
The configuration operation has completed.
Configuration Steps
Log
Writing configuration file
Updating Windows Firewall rules
Adjusting Windows service
Initializing database (may take a long time)
Updating permissions for the data folder and related server files
Starting the server
Applying security settings
Updating the Start menu link
The configuration for MySQL Server 8.0.37 was successful.
Click Finish to continue.
Finish


MySQL Installer
MySQL. Installer
Adding Community
Choosing a Setup Type
Select Products
Installation
Product Configuration
Installation Complete
Product Configuration
We'll now walk through a configuration wizard for each of the following products.
You can cancel at any point if you wish to leave this wizard without configuring all the products.
Product
Status
MySQL Server 8.0.37
Configuration complete.
Next >
Cancel

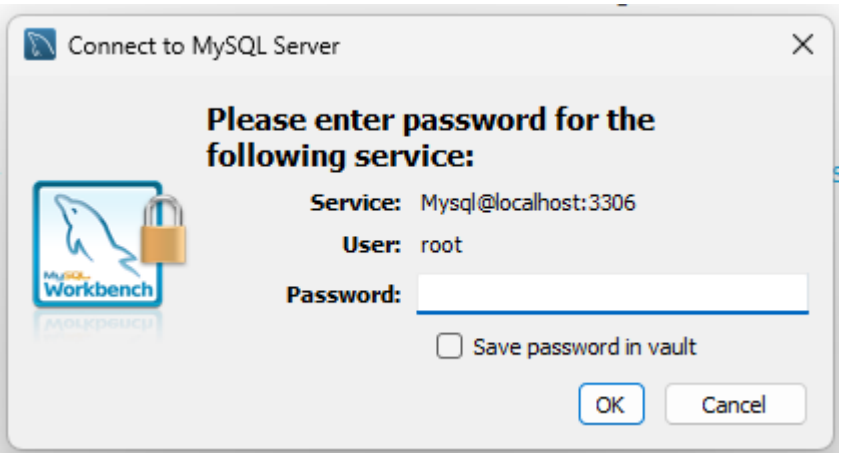


使用 MySQL Workbench 建立範例資料庫

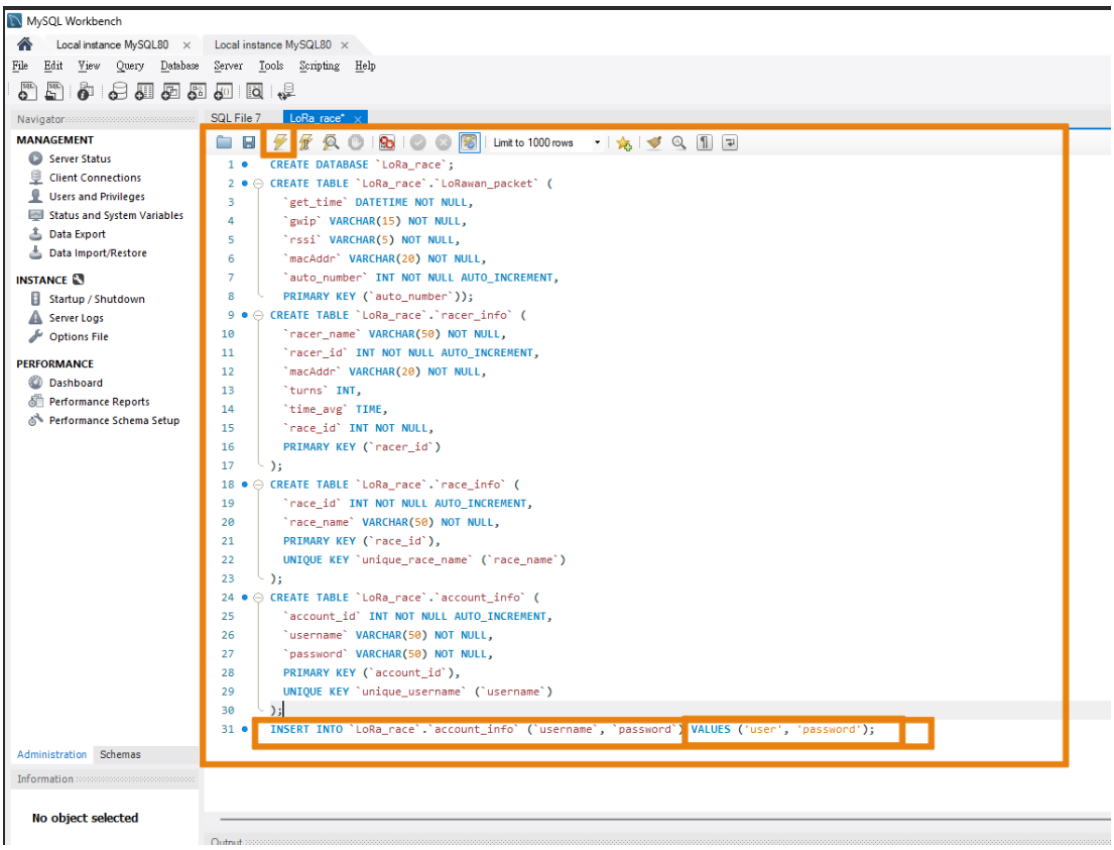
1. 開啟 MySQL Workbench，有依照上面的步驟安裝 MySQL Server 的話，應該會如下圖。接著點擊橘框部分。



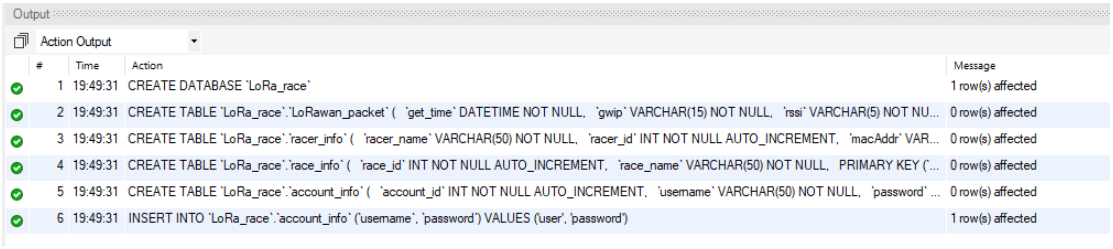
2. 輸入在安裝 MySQL 時設定的 root 密碼。



3. 開啟後將專案根目錄下的 LoRa_race.sql 拖入中間的範圍內，可以修改最後一行的 VALUES('user','password')的內容做為網頁的登入帳號和密碼。修改完成後點擊上方的閃電符號執行 SQL 語法，或是依根目錄下的 ER-Model.png 自行建立資料庫。

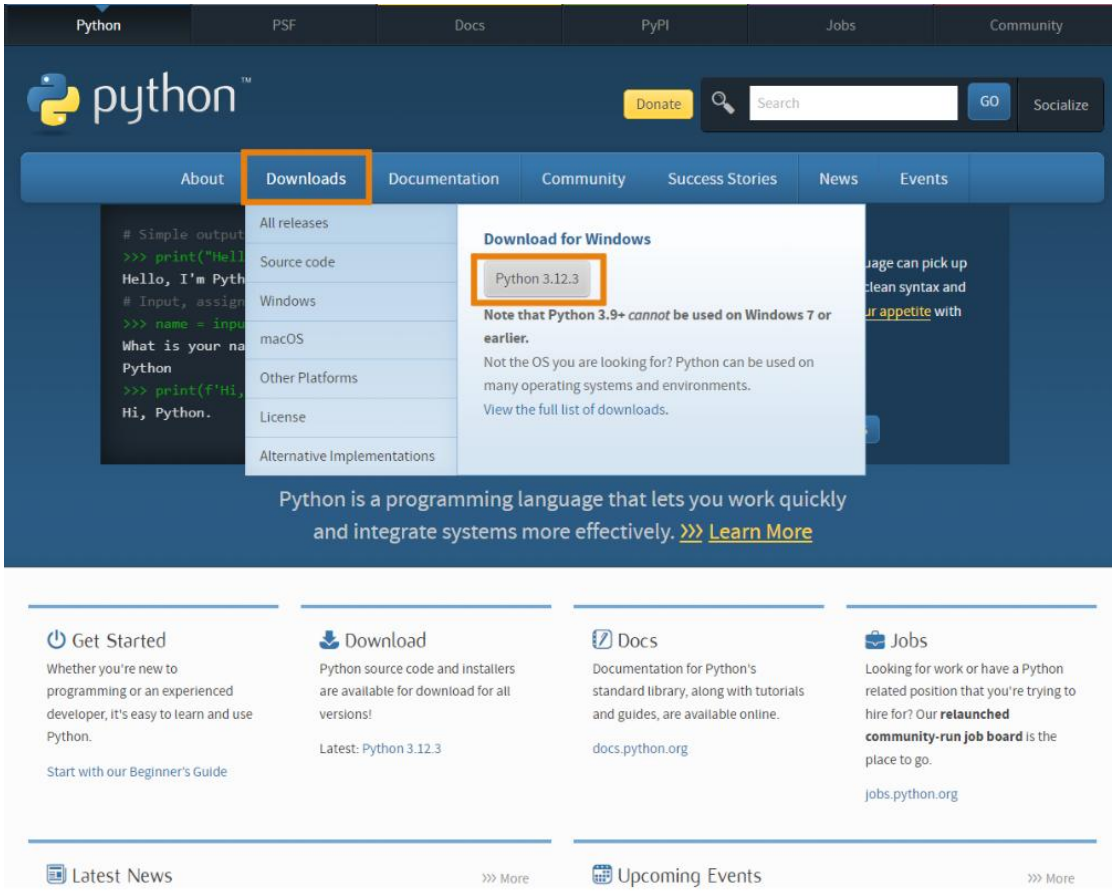


4. 在下方的 Output 若無出現錯誤訊息則成功建立，如下圖。



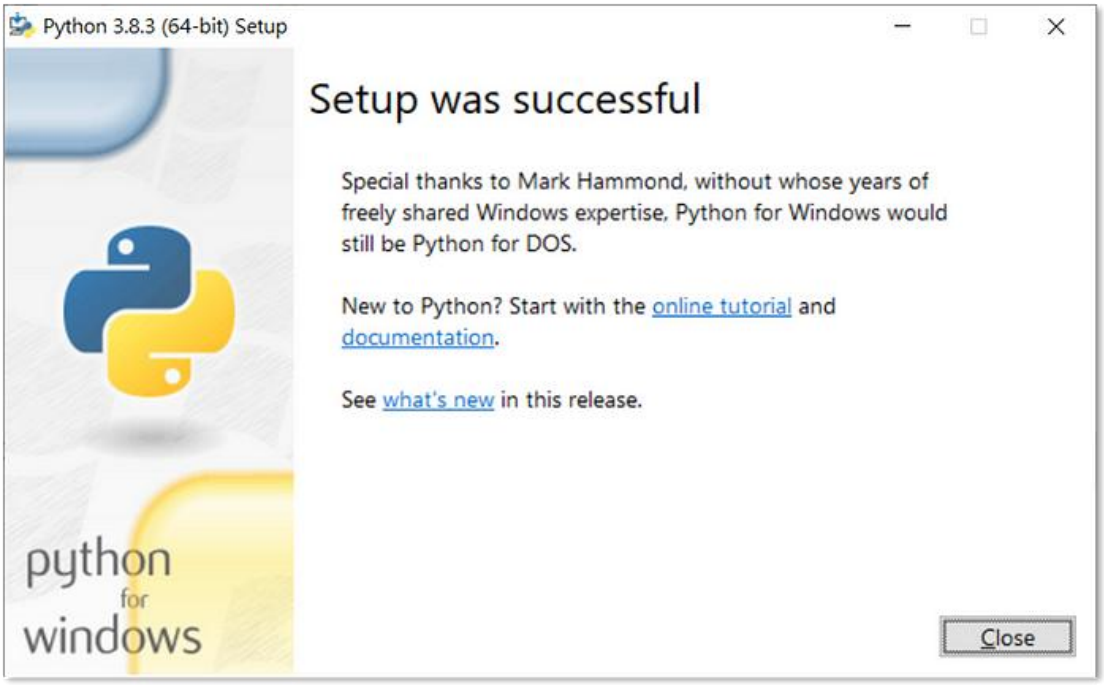
Python 安裝

1. 請前往 python 官網，下載 python，下載最新版即可。



2. 下載完成後，點選安裝檔案後開始安裝，請先勾選下方的 Add python.exe to PATH，勾選後選擇上面的 Install Now 即可。

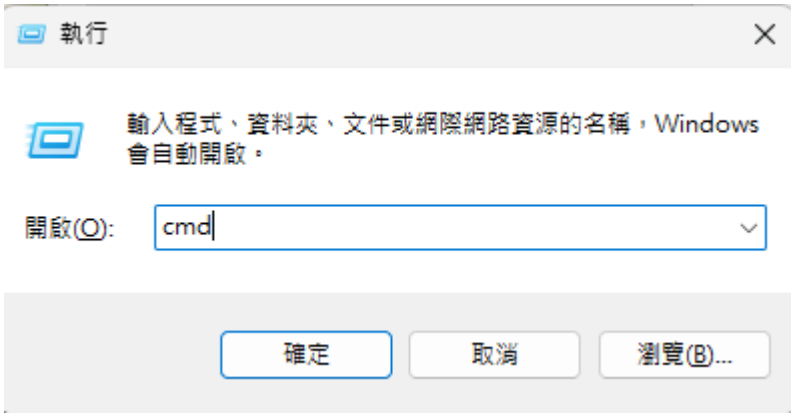
3. 出現下面畫面就表示安裝成功。



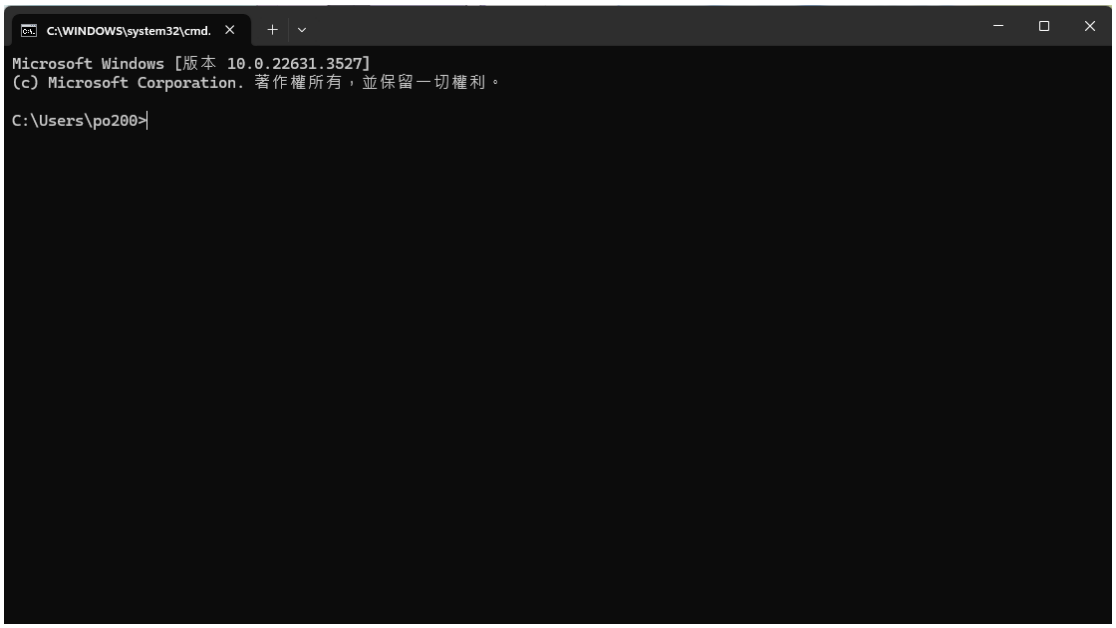
CODING{⌂}LAB

安裝指定 python 套件

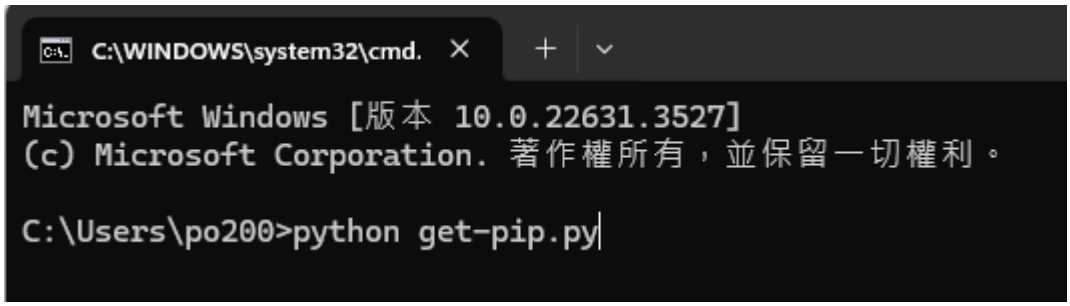
1. 點擊鍵盤上的 Windows 鍵 + R 鍵，會出現下面的視窗。



2. 出現後輸入 cmd 後按下確定，會出現命令提示字元的視窗如下。



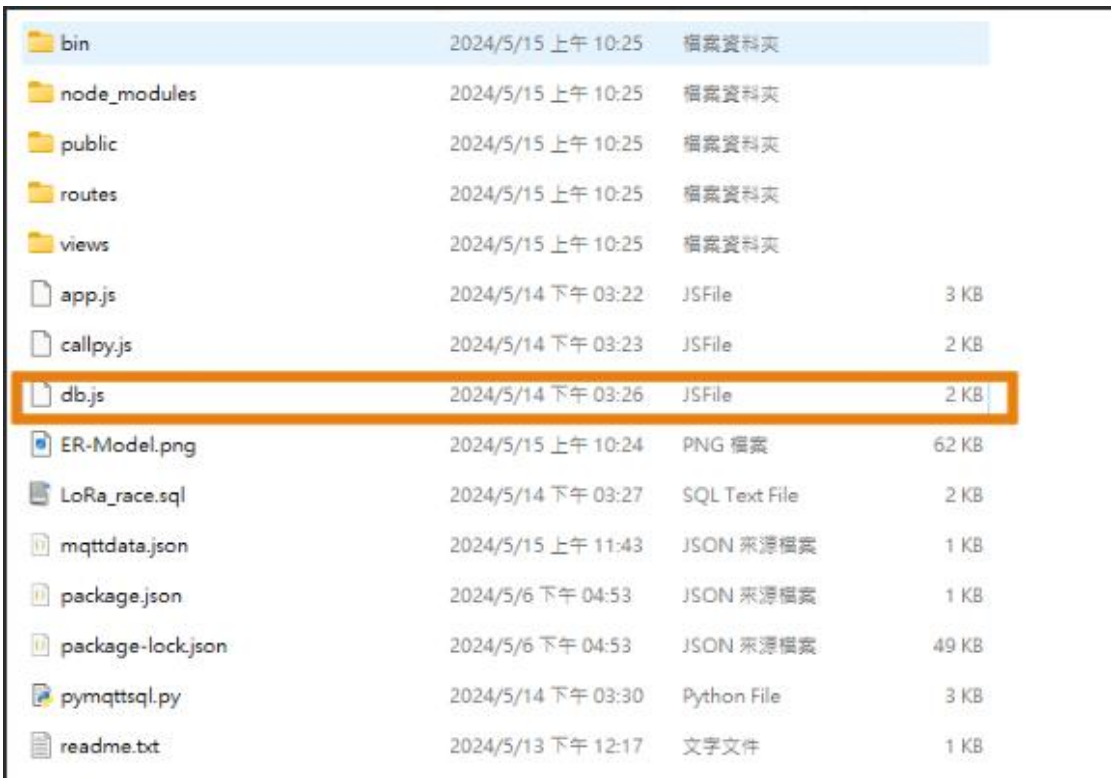
3. 輸入 `python get-pip.py` 確保 pip 是否安裝。



4. 請分次輸入 `pip install paho-mqtt` 和 `pip install pymysql` 來安裝這兩個套件。

網頁連接資料庫設定

1. 開啟本專案根目錄下的 db.js，對該檔案點擊右鍵，在記事本中編輯。



bin	2024/5/15 上午 10:25	檔案資料夾	
node_modules	2024/5/15 上午 10:25	檔案資料夾	
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views	2024/5/15 上午 10:25	檔案資料夾	
app.js	2024/5/14 下午 03:22	JSFile	3 KB
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package.json	2024/5/6 下午 04:53	JSON 來源檔案	1 KB
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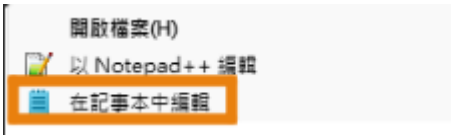
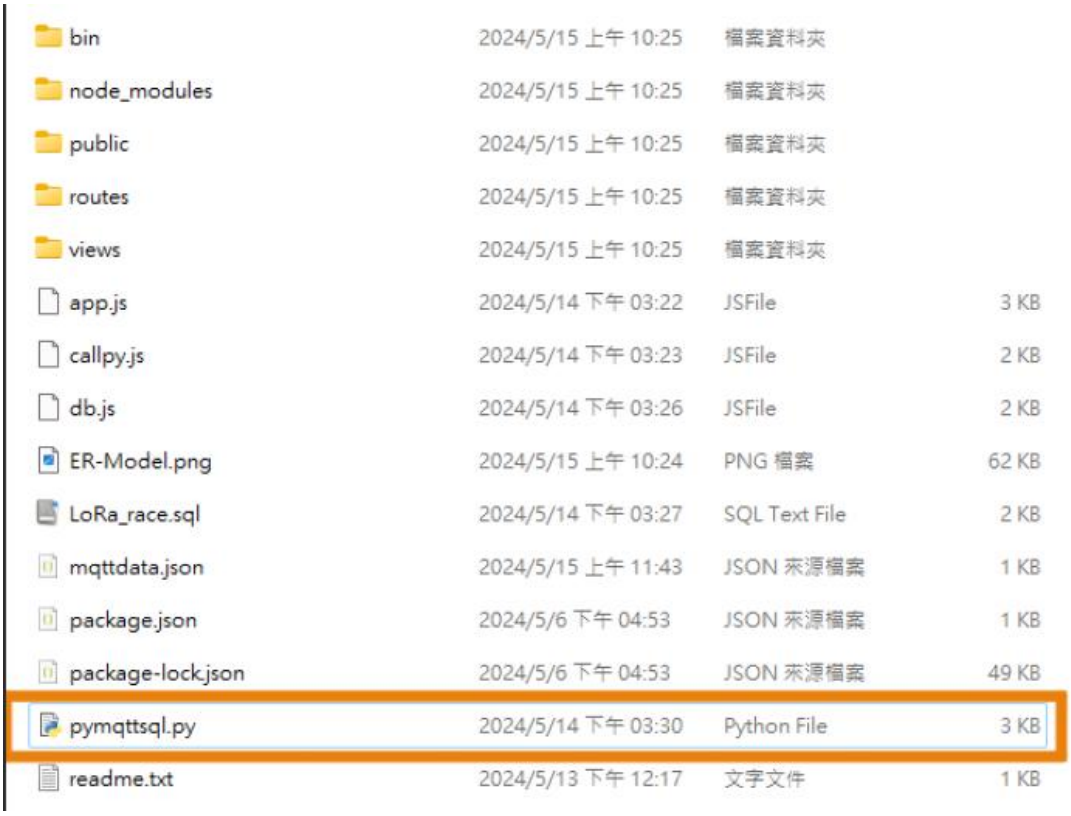
readme.txt	2024/5/13 下午 12:17	文字文件	1 KB



2. 在 fuction connectToMySQL()的函式下的對應部分修改為你的資料庫的設定，修改完後儲存關閉。

```
const mysql = require("mysql");

// 定義函數以連接到 MySQL 資料庫
function connectToMySQL() {
  // 創建 MySQL 連接
  const connection = mysql.createConnection({
    host: "localhost", // 主機名稱
    user: "root", // 使用者名稱
    password: "123456", // 密碼
    database: "LoRa_race", // 資料庫名稱
  });

  // 嘗試與 MySQL 伺服器建立連接
  connection.connect((err) => {
    if (err) {
      // 若連接發生錯誤，輸出錯誤訊息並在 5 秒後重新連線
      console.error("Error connecting to MySQL: " + err.stack);
      setTimeout(connectToMySQL, 5000);
      return;
    }
    // 連接成功，輸出連接執行緒 ID
    console.log("Connected to MySQL as id " + connection.threadId);
  });

  // 監聽 MySQL 連接錯誤事件
  connection.on("error", (err) => {
    console.error("MySQL connection error: " + err.stack);
    // 若連接錯誤，則重新連線
    connectToMySQL();
  });

  // 導出連接物件以便在其他檔案中使用
  module.exports = connection;
}

// 初始連接到 MySQL 資料庫
connectToMySQL();
```

3. 接著開啟本專案根目錄下的 pymqttsql.py，對該檔案點擊右鍵，在記事本中編輯。



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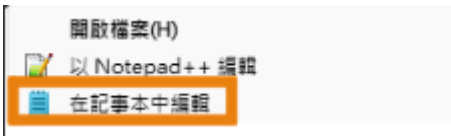

4. 在 def sqlsave (jsonData)下修改為你的資料庫的設定，修改完後儲存關閉。

```
# 接收到消息時的處理函數
def on_message(client, userdata, msg):
    global dddd
    jsondata = json.loads(msg.payload)
    sqlsave(jsondata)

# 主函數
def main():
    # 讀取 MQTT 設置
    config = read_config()
    global topic, host, port, username, password
    host = config.get("hostname")
    port = int(config.get("port"))
    username = config.get("username")
    password = config.get("password")
    topic = config.get("Subscribe_topic")

    # 初始化 MQTT 客戶端
    client = mqtt.Client(protocol=mqtt.MQTTv311, callback_api_version=mqtt.CallbackAPIVersion.VERSION2)
    client.on_connect = on_connect
    client.on_message = on_message
    client.username_pw_set(username, password) # 設置 MQTT 使用者名和密碼
    client.connect(host, port, keepalive=60)
    client.loop_forever()

# 將收到的消息存儲到 MySQL 資料庫中
def sqlsave(jsonData):
    # 連接到 MySQL 資料庫
    db = pymysql.connect(host="localhost", user="root", password="123456", database="lora_race", charset='utf8')
    cursor = db.cursor()
    try:
        # 將 ISO 8601 時間戳轉換為 datetime 對象
        time_iso = jsonData[0]['time']
        time_datetime = datetime.fromisoformat(time_iso)

        # 執行 SQL 插入語句
        sql = "INSERT INTO lorawan_packet (get_time, gwip, rssi, macAddr) VALUES (%s, %s, %s, %s);"
        cursor.execute(sql, (time_datetime, jsonData[0]['gwip'], jsonData[0]['rssi'], jsonData[0]['macAddr']))
        db.commit()
    except Exception as e:
        print("failed")
    finally:
        db.close()

# 讀取 MQTT 設置
def read_config():
    current_dir = os.path.dirname(os.path.abspath(__file__))
    file_path = os.path.join(current_dir, 'mqttdata.json')
    with open(file_path, 'r') as f:
        config = json.load(f)
    return config

# 程式入口
if __name__ == '__main__':
    main()
```

5. 在根目錄下有個 bin 的資料夾，打開後會看到 www 的檔案，一樣對該檔案點擊右鍵，在記事本中編輯。

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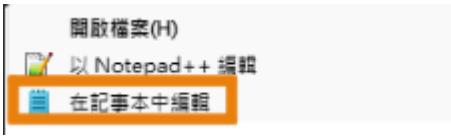
www	2024/4/29 下午 12:20	檔案	2 KB

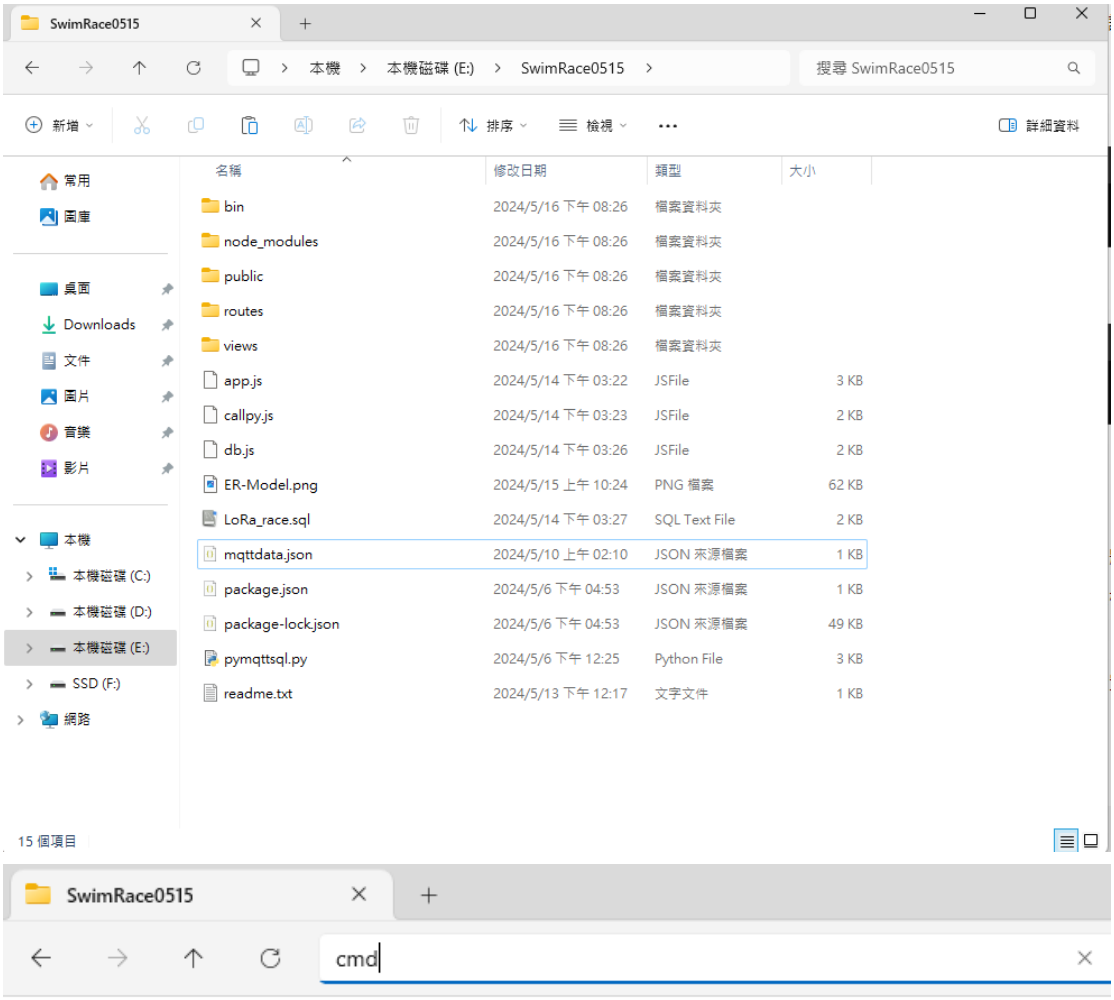


6. 在前幾行會看到 var port = normalizePort(process.env.PORT || '8080');
後面的 8080 代表著該專案啟動時使用的 port，請依自身需求決定是否修改。

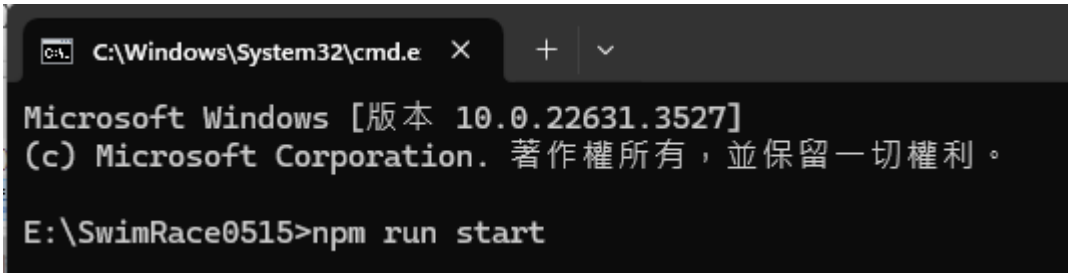
```
#!/usr/bin/env node

/**
 * Module dependencies.
 */

var app = require('../app');
var debug = require('debug')('lora-race:server');
var http = require('http');

/**
 * Get port from environment and store in Express.
 */

var port = normalizePort(process.env.PORT || '8080');
app.set('port', port);
```

啟動 Node.js 伺服器

1. 前往該專案的根目錄下，在路徑上方輸入 cmd，之後 Enter。



2. 輸入 npm run start。



3. 橘框部分代表已經連接到了你的資料庫，此時應該會跳出錯誤訊息如下方的紅框，這是正常的，因為我們還沒設定好 MQTT 的設定。

```
> lora-race@0.0.0 start
> node ./bin/www

Connected to MySQL as id 27
Python 錯誤: Traceback (most recent call last):
  File "E:\SwimRace0515\pymqttsql.py", line 82, in <module>

Python 錯誤:     main()
  File "E:\SwimRace0515\pymqttsql.py", line 51, in main
    client.connect(host,port,keepalive=60)
  File "C:\Users\po200\AppData\Local\Programs\Python\Python310\lib\site-packages\paho\mqtt\client.py", line 1427, in connect
    self.connect_async(host, port, keepalive,

Python 錯誤:   File "C:\Users\po200\AppData\Local\Programs\Python\Python310\lib\site-packages\paho\mqtt\client.py", line 1518, in connect_async
    self.host = host
  File "C:\Users\po200\AppData\Local\Programs\Python\Python310\lib\site-packages\paho\mqtt\client.py", line 891, in host
Python 錯誤:     raise ValueError("Invalid host.")
ValueError: Invalid host.

執行錯誤: Error: Command failed: python pymqttsql.py
Traceback (most recent call last):
  File "E:\SwimRace0515\pymqttsql.py", line 82, in <module>
    main()
  File "E:\SwimRace0515\pymqttsql.py", line 51, in main
    client.connect(host,port,keepalive=60)
  File "C:\Users\po200\AppData\Local\Programs\Python\Python310\lib\site-packages\paho\mqtt\client.py", line 1427, in connect
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    raise ValueError("Invalid host.")
ValueError: Invalid host.
```

4. 此時打開瀏覽器輸入 localhost:8080，後面的 8080 代表 port。
   這時應該就會進到該畫面了。

5. 輸入完帳號密碼後，點選最下面的設備設定。

6. 點選 MQTT 設定

7. 請依照自己的 MQTT 服務器做設定，輸入完畢後點擊”儲存並傳遞資料。

8. 你可以在步驟 2 開啟的 cmd 查看網頁的狀況，你應該會在最下面看到以下訊息。

```
JSON file updated successfully!
POST /mqttpage/receive_data 200 3.867 ms - 49
```

9. 如果你的 MQTT 服務器設定或連線出現問題，則會出現類似下方的錯誤訊息。

```
Python 錯誤: Traceback (most recent call last):
  File "E:\SwimRace0515\pymqttsql.py", line 82, in <module>

Python 錯誤:     main()
  File "E:\SwimRace0515\pymqttsql.py", line 51, in main
    client.connect(host,port,keepalive=60)
  File "C:\Users\po200\AppData\Local\Programs\Python\Python310\lib\site-packages\paho\mqtt\client.py", line 1427, in connect

Python 錯誤:     self.connect_async(host, port, keepalive,
  File "C:\Users\po200\AppData\Local\Programs\Python\Python310\lib\site-packages\paho\mqtt\client.py", line 1518, in connect_async

Python 錯誤:     self.host = host
  File "C:\Users\po200\AppData\Local\Programs\Python\Python310\lib\site-packages\paho\mqtt\client.py", line 891, in host

Python 錯誤:     raise ValueError("Invalid host.")
ValueError: Invalid host.

執行錯誤: Error: Command failed: python pymqttsql.py
Traceback (most recent call last):
  File "E:\SwimRace0515\pymqttsql.py", line 82, in <module>
    main()
  File "E:\SwimRace0515\pymqttsql.py", line 51, in main
    client.connect(host,port,keepalive=60)
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ValueError: Invalid host.
```

10. 若要關閉網頁的話，關閉步驟 1 和 2 所執行的 cmd 即可。